

Drift Engine Teaching Workbook

A Multi-Session Practitioner Curriculum in Drift Detection, Calibration & Dimensional Development

Version: R5.1 **Module:** RTT_DRIFT_ENGINE **Path:** clarity/models/drift

Sessions: 8 **Level:** Advanced Practitioner

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Workbook Preface & How to Use This Guide

Why Drift Matters

Drift is the foundational failure mode of any intelligent evaluation system. Before a system hallucinates, before it loses intent coherence, before it produces corrupted output — it drifts. Drift is not an endpoint; it is a trajectory. The `RTT_DRIFT_ENGINE` model at `clarity/models/drift` exists precisely to detect, measure, and correct that trajectory before it becomes irreversible. Every practitioner working within the TriadicFrameworks R5 canon must understand drift not merely as a concept, but as a phenomenon they can read in real time, score quantitatively, and correct through structured intervention.

This workbook exists because drift literacy is non-trivial. It requires simultaneous fluency in substrate dynamics, pipeline mechanics, dimensional analysis, and metacognitive calibration. No single session can convey this. A progressive, layered curriculum is required — one that builds each competency on the last, tests understanding before advancing, and synthesizes all layers into a unified evaluative practice.

R5 Canon Pedagogy: The Five Pillars of R5 Learning

All TriadicFrameworks R5 pedagogical materials are organized around five learning pillars:

Pillar	Name	Description
R1	Recognize	Identify the phenomenon in context. Before you can act, you must perceive accurately.
R2	Reflect	Analyze the phenomenon without premature judgment. What is actually happening?
R3	Reframe	Contextualize the phenomenon within the appropriate dimensional or systemic lens.
R4	Realign	Apply the appropriate corrective action. Bring the system back into calibrated operation.
R5	Resolve	Confirm resolution, document findings, and prevent recurrence through calibration updates.

Each session of this workbook is aligned to one or more of these pillars. Sessions 1–2 are primarily R1/R2 (Recognize, Reflect). Sessions 3–4 develop R2/R3 competency. Sessions 5–6 build R4 (Realign). Session 7 applies R5 (Resolve) in evaluator mode. Session 8 integrates all five pillars in capstone synthesis.

Session Architecture

Every session in this workbook follows a consistent five-part structure:

1. **Learning Objective** — A single, precise, measurable statement of what the practitioner will be able to do upon completing the session.
2. **Concept Exposition** — Structured conceptual content, presented with technical precision and organized into subsections with supporting tables, diagrams, and formulas.

3. **Worked Example** — A fully annotated, step-by-step application of session concepts to a realistic drift scenario. Practitioners should read worked examples actively, not passively.
4. **Practice Tasks** — Hands-on exercises requiring the practitioner to apply session concepts independently. Answer spaces are provided.
5. **Clarity Pulse Checkpoint** — A structured, brief knowledge-check covering session essentials. Serves as both self-assessment and readiness gate before advancing.

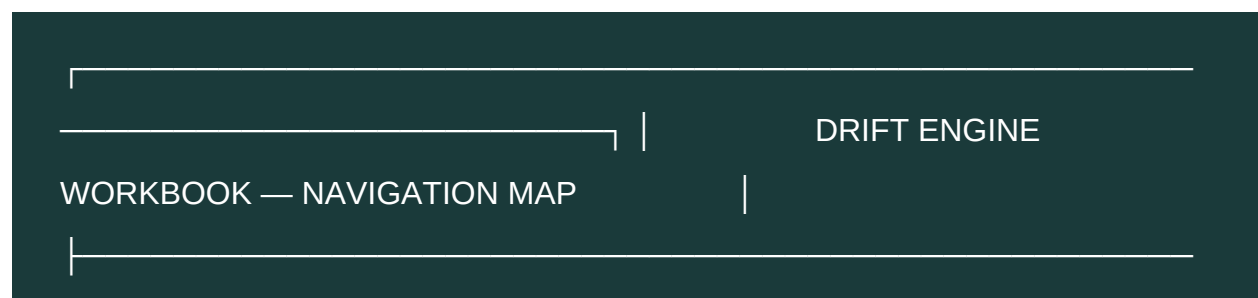
Practitioner Navigation Notes

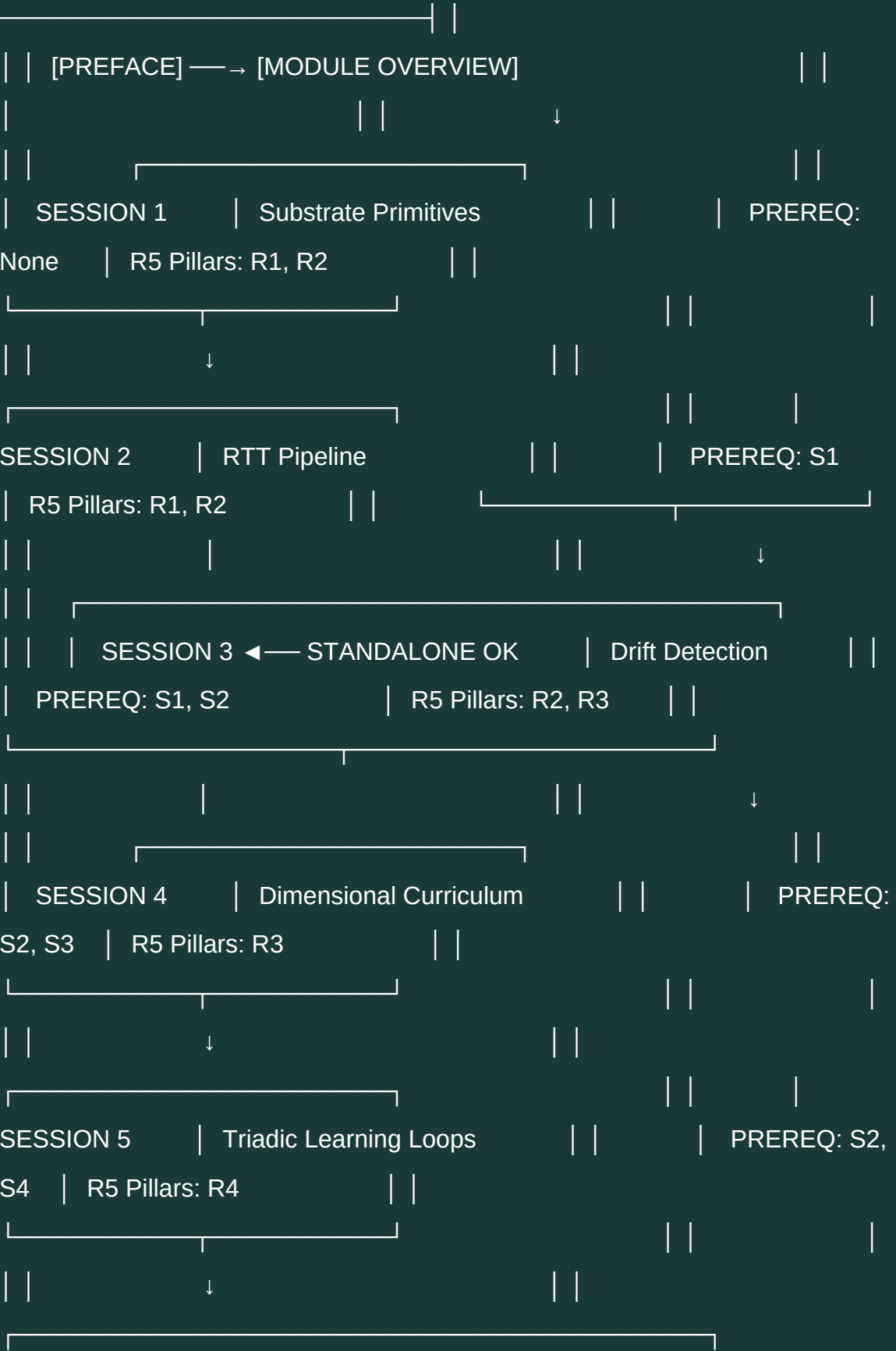
First-pass practitioners: Complete sessions 1 through 8 in sequence. Each session builds directly on the prior. Skipping sessions on first pass will produce gaps that surface painfully in Sessions 7 and 8.

Advanced practitioners: Sessions 3, 6, and 7 may be used as standalone reference materials for threshold calibration, CPC deployment, and evaluator exercise respectively. The appendices are designed for independent lookup at any time.

All practitioners: Do not skip Clarity Pulse Checkpoints. They are not optional review — they are calibration gates that confirm your understanding is accurate before you proceed. A wrong answer at a CPC is data, not failure.

Workbook Navigation Map





Canonical Module Fields

The following table presents the complete canonical field set for the RTT_DRIFT_ENGINE module, as defined in the R5.1.0 release of the TriadicFrameworks canon. Practitioners should treat this table as a primary reference document alongside the full Appendix A field listing.

Field	Type	Default / Value	Description
module_id	string	RTT_DRIFT_ENGINE	Unique module identifier within the TriadicFrameworks registry. Used for cross-module references and pipeline routing.
version	semver	R5.1.0	Semantic version of the module release. Major version tracks canon generation (R5); minor tracks feature releases; patch tracks hotfixes.
substrate	object	See Session 1	Defines the four primitive substrate layers: signal, noise, flux, anchor. Each layer has sub-fields for entropy index, flux index, and type classification.
pipeline_stages	array	[Intake, Parse, Evaluate, Resolve, Emit]	Ordered array of the five RTT pipeline stage identifiers. Stages execute in sequence; each stage's output feeds the next as input.
drift_threshold	float	0.38	Default drift sensitivity threshold. Parse units scoring above this value are flagged as Drift Candidates. Adjustable via calibration_curve parameters.

Field	Type	Default / Value	Description
<code>dimensional_axes</code>	array	[D1, D2, D3, D4, D5]	Five named dimensional axes used to contextualize drift signals: Temporal (D1), Semantic (D2), Relational (D3), Intentional (D4), Integrative (D5).
<code>triadic_loop</code>	object	See Session 5	Configuration object for the Perceive-Process-Project correction cycle. Contains phase durations, re-entry conditions, and RIS weighting parameters.
<code>pulse_interval</code>	int	8	Clarity Pulse Checkpoint cadence in evaluation ticks. At every 8th tick, a CPC is automatically triggered regardless of current DS values.
<code>evaluator_mode</code>	enum	<code>passive</code>	Operational mode of the human evaluator interface. Values: <code>passive</code> (observe only), <code>active</code> (verify and validate), <code>corrective</code> (direct intervention).
<code>calibration_curve</code>	object	$k=2.0$, $x_0=0.38$	Sigmoid calibration curve parameters. k controls steepness (sensitivity); x_0 sets the inflection point (baseline threshold).
<code>emit_schema</code>	object	See Session 2	Output schema definition for resolved drift signals. Specifies fields for corrected substrate object, <code>drift_log</code> entry, and optional pulse signal.

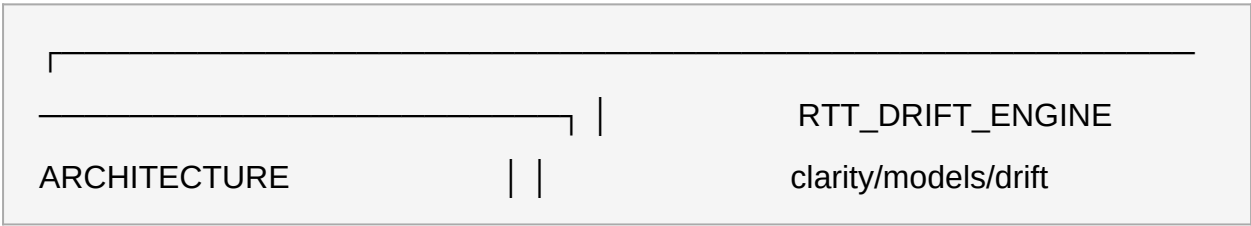
How the Fields Work Together

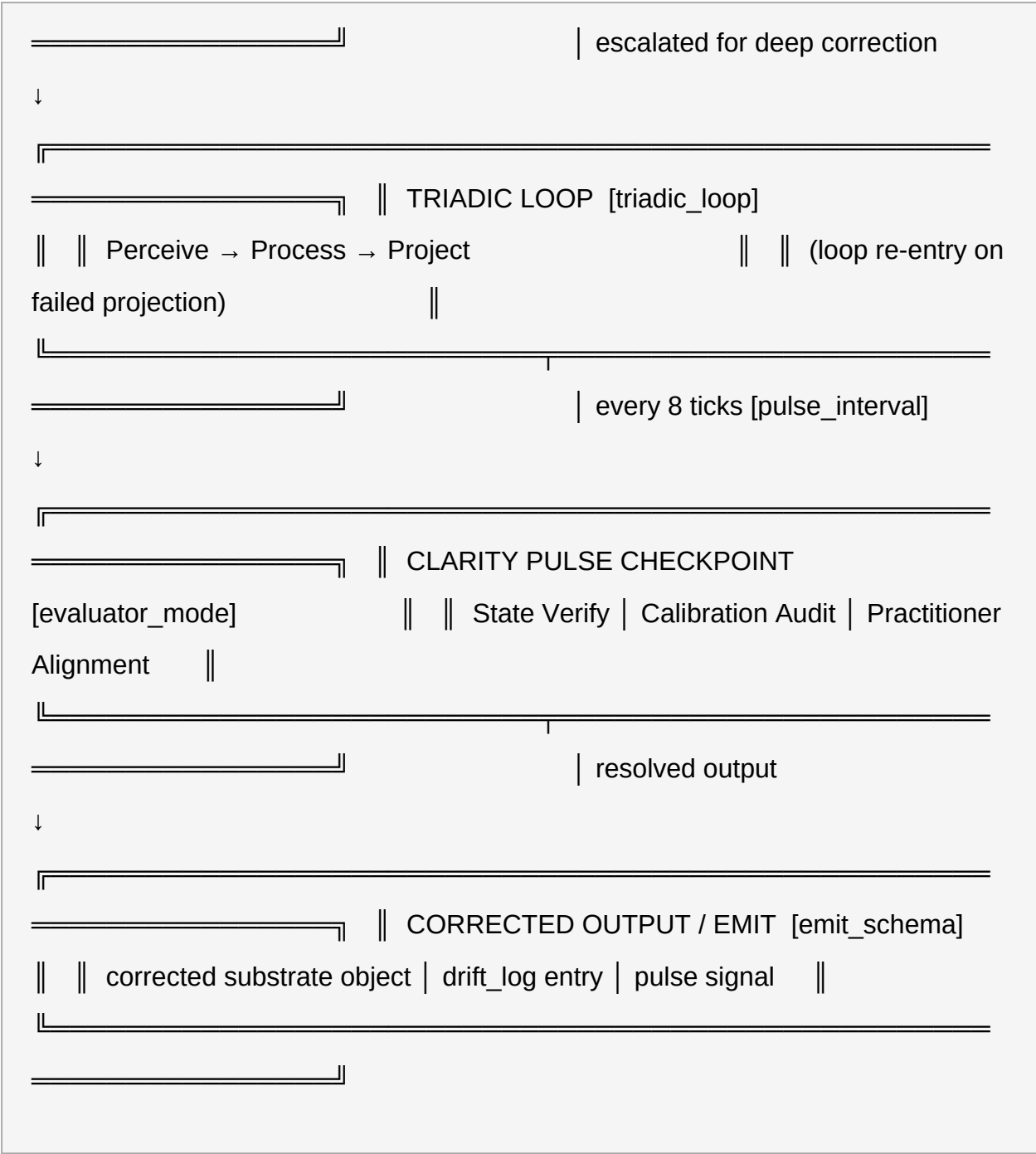
The `RTT_DRIFT_ENGINE` fields are not independent configuration parameters — they form a unified operational architecture. Understanding their interdependencies is essential to effective drift evaluation. The substrate object is the engine's raw material: it defines what is being evaluated and in what condition it arrives. The `pipeline_stages` array defines the transformation path that substrate travels as it is analyzed. At the Evaluate stage, the `drift_threshold` determines which parse units raise flags, and the `calibration_curve` modulates that threshold dynamically based on observed variance history.

Once drift candidates are identified, they are contextualized through the `dimensional_axes`. Each axis provides a different diagnostic lens — a single flagged parse unit might score high on D2 (semantic deviation) but show no D1 (temporal) anomaly, suggesting that the drift is content-level rather than structural. This dimensional triage is not automatic; it requires practitioner interpretation guided by the rubrics in Session 4. When dimensional analysis indicates the drift is too complex for simple recalibration or suppression, the `triadic_loop` object activates the Perceive-Process-Project cycle, providing a structured framework for deeper corrective intelligence.

Throughout the evaluation, the `pulse_interval` field ensures that the Clarity Pulse Checkpoint system fires at regular intervals, preventing evaluator drift — the phenomenon where the human operator gradually loses alignment with the engine's assessment. The `evaluator_mode` determines how much authority the human evaluator has to intervene. All outputs are structured by the `emit_schema`, ensuring that regardless of resolution path, the output is consistently formatted for downstream systems and for the `drift_log` that supports recurrence analysis.

Architecture Diagram





Learning Objective

Identify and distinguish the four substrate primitives — Signal, Noise, Flux, and Anchor — and explain how each conditions the drift signal in the RTT_DRIFT_ENGINE evaluation environment.

Concept Exposition

1. Signal — The Raw Informational Carrier

Signal is the foundational primitive. It is defined as structured input that carries intent-bearing content — information that has a discernible purpose, direction, or semantic load. Signal is what the Drift Engine is ultimately trying to evaluate: the substrate of meaning. Without a legible signal, the entire pipeline has nothing to work with.

Signal health is measured by its **entropy index** (EI). A signal is considered stable when $EI < 0.30$. As entropy rises above 0.30, signal begins to lose its structural integrity — the information becomes less organized, less predictable, and more difficult to parse into coherent units. At $EI > 0.60$, a signal is classified as severely degraded and may not be parseable without preliminary remediation.

Signal purity refers to the proportion of a signal's content that is genuinely intent-bearing versus incidentally present. A pure signal (purity > 0.80) contains almost entirely meaningful content. **Signal contamination** occurs when noise elements infiltrate the signal structure, reducing purity and introducing parse errors. Contamination is often the first detectable sign of upstream drift.

Signal Stability Threshold

$EI < 0.30 \rightarrow$ Stable Signal | $EI \geq 0.30 \rightarrow$ Degraded Signal | $EI \geq 0.60 \rightarrow$ Severely Degraded

2. Noise — Undifferentiated Interference

Noise is the adversarial primitive. It occupies the same substrate space as signal but carries no intent-bearing content. Noise exists on a spectrum from **thermal noise** (fully random, statistically uniform, with no internal structure) to **structured noise** (patterned interference that mimics the surface features of a legitimate signal). The distinction is critical: thermal noise is relatively easy to filter because it has no semantic coherence, no repetition pattern, and no consistent frequency signature. Structured noise is far more dangerous precisely because it looks like signal.

The **Noise Differentiation Protocol (NDP)** provides the standard methodology for distinguishing structured noise from legitimate signal. The NDP compares three features:

6. **Frequency Signature:** Does the input exhibit a consistent frequency pattern? Thermal noise has no dominant frequency; structured noise often does.
7. **Repetition Interval:** How frequently do similar units recur? Signal repetition follows semantic logic; noise repetition follows interference logic (shorter, more mechanical intervals).
8. **Semantic Coherence:** Does the input maintain meaning across adjacent parse units? Genuine signal has semantic continuity; structured noise may produce locally coherent fragments but loses coherence at the relational level.

Practitioner Note: Never attempt to filter structured noise using the same tools appropriate for thermal noise. Applying a simple frequency filter to structured noise will eliminate it in the short term but cause the engine to lose calibration against its interference pattern. The correct response to structured noise is NDP classification followed by anchor assignment.

3. Flux — The Rate of Change Across Substrate

Flux is the dynamic primitive. It measures the rate of change across the substrate field — how rapidly the substrate's characteristics are shifting over time. Flux is not inherently negative; a substrate can change while remaining coherent and signal-bearing. Low-to-moderate flux is normal in any active evaluation environment. Flux becomes a concern when it exceeds the system's ability to track and anchor the changing substrate.

The **Flux Index (FI)** provides a quantitative measure of substrate velocity:

Flux Index Formula

$$FI = \Delta \text{signal} / \Delta \text{time}$$

Where Δsignal represents the magnitude of substrate change (measured as the absolute difference in signal coherence score) and Δtime is measured in evaluation ticks. A Flux Index above 0.60 indicates an unstable environment where drift likelihood increases nonlinearly — small additional changes produce disproportionately large drift effects. At $FI > 0.80$, automatic anchor assignment protocols should be triggered.

4. Anchor — Fixed Reference Points Against Flux

Anchors are the stabilizing primitive. They are substrate elements — manually or automatically assigned — that resist flux and provide fixed reference points against which drifting substrate can be measured and corrected. Without anchors, a high-flux environment has no stable ground: there is nothing to drift *from*, and therefore nothing to calibrate back *to*.

The RTT_DRIFT_ENGINE recognizes three types of anchors:

- **Temporal Anchors:** Fixed time-indexed reference points. Prevent temporal drift by maintaining a stable chronological frame against which sequence anomalies can be detected.

- **Semantic Anchors:** Fixed meaning-reference points. Prevent semantic drift by maintaining a stable intent vector baseline against which content deviation can be measured.
- **Relational Anchors:** Fixed relational reference points. Prevent relational drift by maintaining a stable map of how parse units connect to each other, providing resistance against orphaning and circular pattern formation.

Substrate Interaction Matrix

Primitive ×	Signal	Noise	Flux	Anchor
Signal	Self-reinforcing. High signal density increases parse reliability and suppresses noise infiltration.	Contaminati on Risk. Noise infiltrates signal structure; reduces purity; produces parse errors.	Velocity Effect. Flux changes signal characteristics over time; requires continuous re-evaluation.	Stability Zone. Anchors lock signal reference, providing a stable baseline for drift measurement.
Noise	Contaminati on Risk. See Signal × Noise.	Self-amplifying in structured form. Thermal noise is stable but always-present background.	Critical Drift Zone. High noise + high flux = maximum drift risk. NDP becomes unreliable; anchors mandatory.	Isolation Effect. Anchors prevent noise from infiltrating anchor-adjacent signal zones.
Flux	Velocity Effect. See Signal × Flux.	Critical Drift Zone. See Noise × Flux.	Self-accelerating above FI=0.60. Flux generates additional flux as the substrate destabilizes.	Dampening Effect. Anchors reduce effective flux in their vicinity, creating local stability zones.
Anchor	Stability Zone. See Signal × Anchor.	Isolation Effect. See Noise × Anchor.	Dampening Effect. See Flux × Anchor.	Anchor-on-anchor: redundancy. Multiple anchor types in the same zone create over-constrained substrate — valid but may

Primitive ×	Signal	Noise	Flux	Anchor
				suppress legitimate signal variance.

Worked Example 1 — Noise Differentiation Protocol in Action

Scenario: A practitioner observes an input stream where semantic coherence drops from 0.78 to 0.41 over 12 ticks, while repetition interval tightens from 8 to 3 ticks. Is this thermal noise, structured noise, or something else?

Step 1 — Frequency Signature Check: The coherence drop is monotonic (not random), which suggests a pattern. Thermal noise would produce irregular, non-directional coherence variance. A sustained directional drop over 12 ticks is a frequency signature indicator of structured interference.

Step 2 — Repetition Interval Check: Repetition interval narrowing from 8 to 3 ticks indicates acceleration — the interference is becoming more frequent and more intense. This is a hallmark of structured noise, not thermal noise. Thermal noise repetition intervals are random; structured noise intervals are mechanical and typically tightening.

Step 3 — Semantic Coherence Check: The drop from 0.78 to 0.41 crosses the 0.45 semantic coherence threshold (introduced in Session 4 as the D2 drift indicator). Coherence at 0.41 means the substrate is failing to maintain meaning continuity — but the pattern of the drop (gradual, directional) suggests a structural cause rather than random signal degradation.

NDP Conclusion: This is **structured noise with high flux**. $FI = \Delta\text{signal} / \Delta\text{time} = 0.37 / 12 = 0.031$ per tick, which seems low — but the repetition interval tightening indicates local flux acceleration that the global FI does not capture. A localized flux analysis is warranted. **Anchor assignment is required immediately** — specifically, a semantic anchor to hold the coherence baseline, and a temporal anchor to resist further interval compression.

Practice Task 1.1 — Substrate Scenario Classification

For each of the three substrate scenarios below, identify the **dominant primitive at play** and the **recommended first response**.

Scenario A: Input entropy index = 0.18. No repetition anomalies. Flux index = 0.12. Signal purity estimated at 0.87. All parameters stable over 20 ticks.

Dominant Primitive:

Recommended First Response:

Scenario B: Input shows random coherence fluctuations (± 0.15) with no discernible pattern. Repetition intervals vary from 2 to 19 ticks with no regularity. FI = 0.08. No drift candidates flagged.

Dominant Primitive:

Recommended First Response:

Scenario C: Semantic coherence was 0.82 at tick 1, now 0.51 at tick 18. Three temporal anchors have been auto-assigned. FI = 0.73 and rising.

Dominant Primitive:

Recommended First Response:

Practice Task 1.2 — Flux Index Calculation

Given: $\Delta\text{signal} = 0.34$, $\Delta\text{time} = 7$ ticks.

Calculate the Flux Index (FI) and interpret the result. Is this a concern? What action, if any, is recommended?

Calculation:

Result: FI =

Interpretation (write here):

Recommended Action:

📍 Clarity Pulse Checkpoint — Session 1

Q1. Name the four substrate primitives.

Q2. What distinguishes structured noise from thermal noise? Identify two distinguishing features.

Q3. When does high flux become a critical concern? State the threshold value.

Q4. What are the three types of anchors?

The RTT Pipeline: Reading Drift as Signal

Learning Objective

Trace a drift signal through all five RTT pipeline stages and correctly annotate what happens at each stage — including inputs, transformations, outputs, and decision points.

Concept Exposition — Five Pipeline Stages

Stage 1: Intake

The Intake stage is the pipeline's entry point. Raw substrate arrives here from the external environment and undergoes initial substrate typing — the engine's first-pass classification of what it is receiving. At Intake, the engine classifies the incoming substrate as one of four types: Signal-dominant, Noise-dominant, Flux-dominant, or Anchor-bearing. This classification is not final — it is a working hypothesis that subsequent stages will refine — but it determines which parsing heuristics are applied downstream.

The Intake stage produces the **Intake Manifest**, a structured record that all subsequent stages reference. The Intake Manifest captures four fields:

Intake Manifest Field	Description	Example Value
substrate_type	Initial classification: Signal, Noise, Flux-dominant, or Anchor-bearing	Signal
entropy_index	Measured entropy of the incoming substrate at time of intake	0.24
flux_index	Flux Index at intake, calculated from recent delta	0.41

Intake Manifest Field	Description	Example Value
	history	
timestamp	Tick count at which intake occurred	tick_07

Stage 2: Parse

The Parse stage decomposes the substrate into **parse units** — the smallest semantically coherent fragments of the substrate that can be evaluated independently. A parse unit is not merely a syntactic fragment; it must carry at least a partial intent vector to qualify. Sub-threshold fragments are treated as parse debris and logged but not evaluated.

Parsing produces a **parse tree**: a hierarchical representation of how parse units relate to each other. The parse tree is the data structure that all downstream stages operate on. Parse depth varies by substrate complexity:

- **Shallow Parse (depth 1):** Surface features only — unit type, length, entropy, direct adjacency relationships.
- **Mid Parse (depth 2):** Relational patterns — how units connect to each other; referential chains; semantic groupings.
- **Deep Parse (depth 3):** Intent vectors — the underlying purpose or direction of each parse unit; goal-frame analysis.

Stage 3: Evaluate

The Evaluate stage is the core of the pipeline. Each parse unit produced by the Parse stage is scored against the `drift_threshold` using the **Drift Score (DS) formula**:

Drift Score Formula

$$DS = (\text{semantic_deviation} \times \text{flux_weight}) / \text{anchor_presence}$$

Where **semantic_deviation** is the measured departure of the parse unit's content from the established intent vector baseline (range 0–1), **flux_weight** is the current Flux Index applied as a multiplier to amplify drift scores in high-volatility environments, and **anchor_presence** is a value from 0.1 to 1.0 representing the density of anchors in the parse unit's vicinity. Anchor presence in the denominator is the critical design decision: it means that well-anchored parse units require much higher semantic deviation to trigger a drift flag, reducing false positives in stable substrate zones.

Parse units scoring $DS > 0.38$ (drift_threshold) are flagged as **Drift Candidates** and passed to the Resolve stage. Units scoring below threshold are marked as Clean and included in the corrected substrate output.

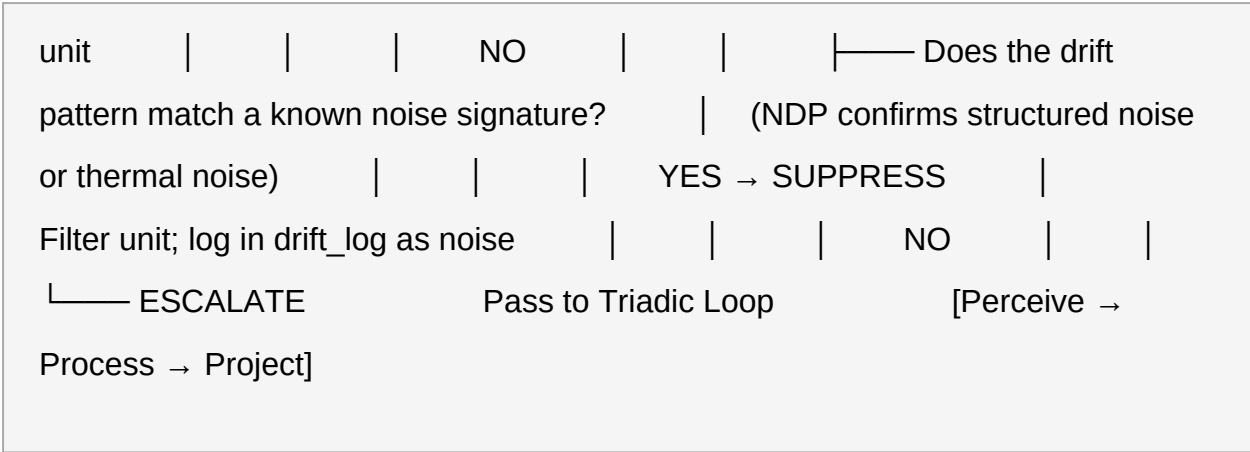
Stage 4: Resolve

The Resolve stage is the decision stage. Flagged Drift Candidates arrive here for disposition. The engine applies one of three resolution strategies:

9. **Recalibrate:** Adjust the drift_threshold for this substrate stream. Used when the drift pattern suggests the baseline threshold is misconfigured for this environment — the drift is not genuine deviation, but threshold error.
10. **Suppress:** Filter the drifting parse unit as noise. Used when the unit's drift pattern matches known noise signatures and the unit contributes no recovery value if retained.
11. **Escalate:** Pass the Drift Candidate to the triadic loop for deeper corrective processing. Used when DS is high, anchor presence is low, and simple threshold adjustment or suppression is insufficient.

RESOLVE STAGE — DECISION TREE Drift Candidate arrives ($DS > 0.38$)

	—— Is $DS > 0.38$ due to threshold miscalibration?		(Historical
DS variance suggests this environment		systematically scores above	
baseline threshold)		YES → RECALIBRATE	
Adjust calibration_curve parameters		for this stream; re-evaluate	



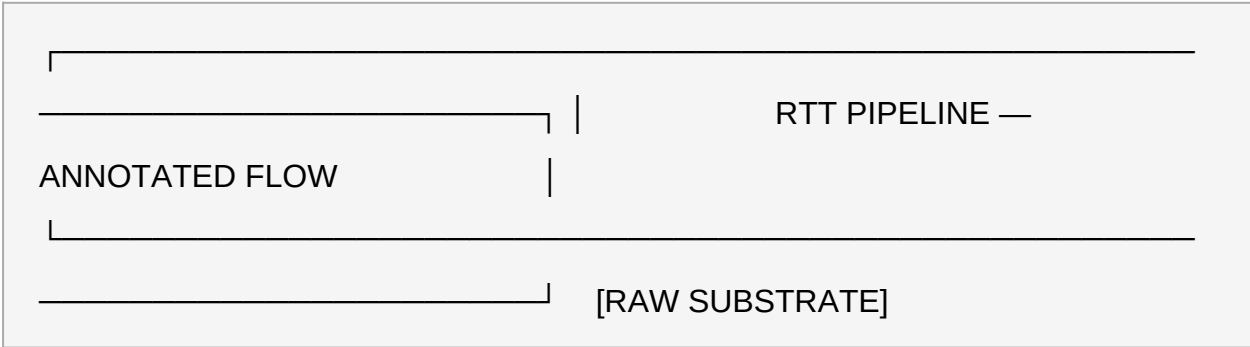
Stage 5: Emit

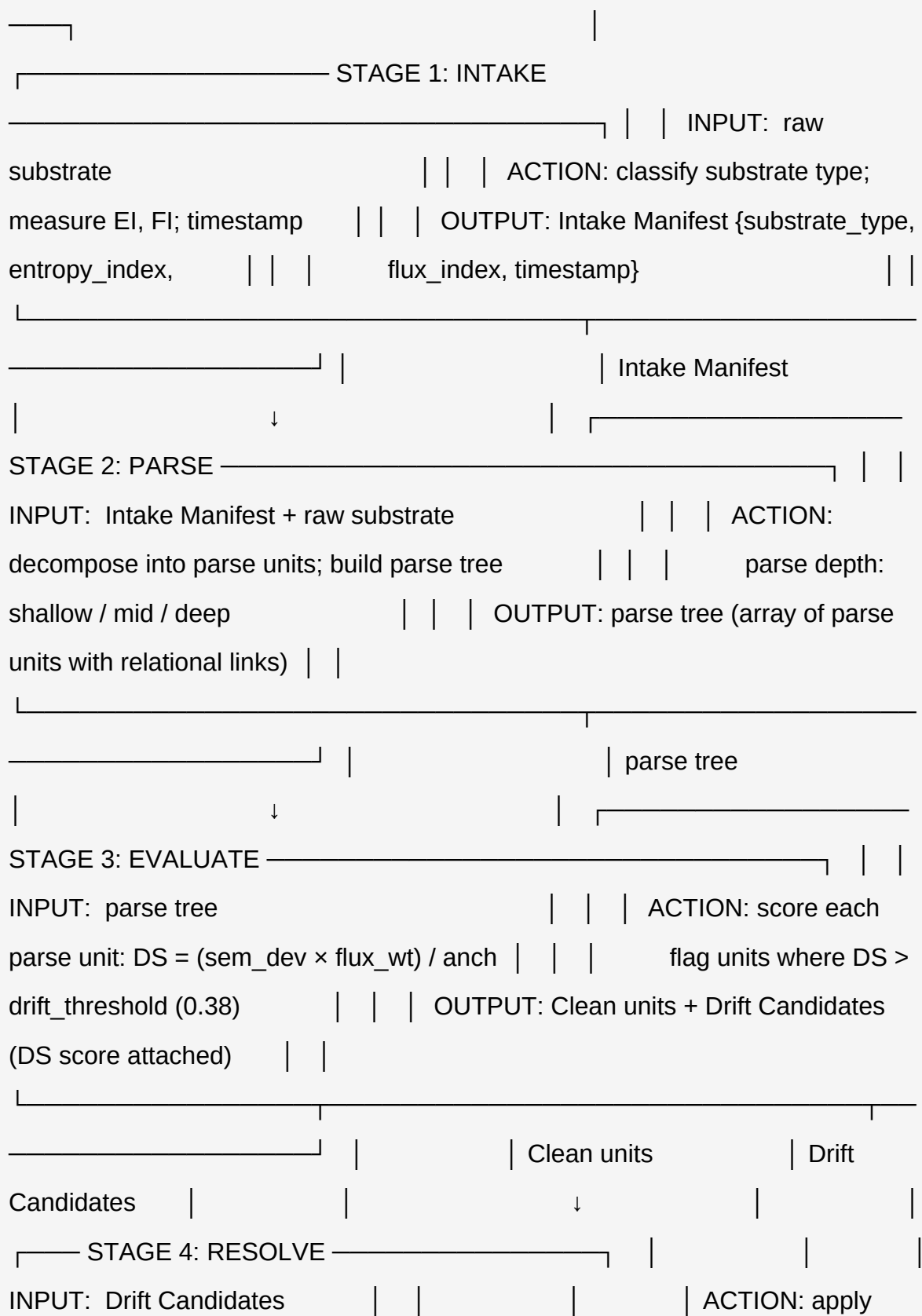
The Emit stage packages resolved output per the emit_schema. Regardless of which resolution strategy was applied, every Emit cycle produces three outputs:

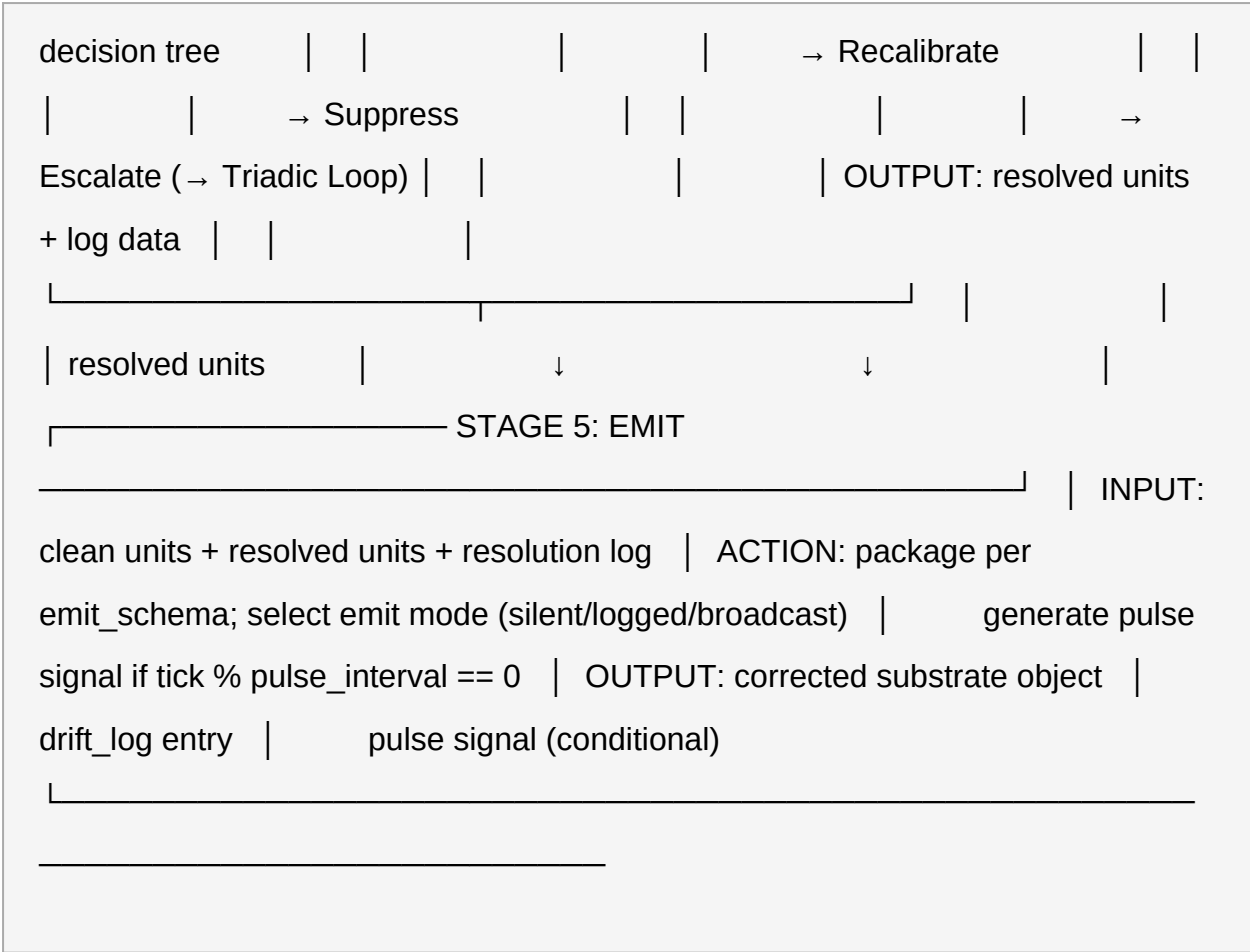
- 12. **Corrected Substrate Object:** The substrate with drifting units resolved (recalibrated, suppressed, or replaced by triadic loop output).
- 13. **Drift Log Entry:** A structured record of the drift event, including DS, dimensional flags, resolution strategy applied, and outcome.
- 14. **Pulse Signal (conditional):** If a Clarity Pulse Checkpoint is scheduled at this tick, the Emit stage generates a pulse signal for the CPC system.

Emit operates in three modes: **silent** (outputs stored internally, not surfaced), **logged** (outputs written to drift_log; corrected substrate returned), and **broadcast** (all outputs surfaced to external systems, evaluator notified).

Pipeline Flow Diagram







Worked Example 2 — Full Pipeline Trace: Semantic Frame Shift

Input Event: "Respondent shifts semantic frame mid-stream from temporal to causal reasoning without anchor. Semantic coherence drops from 0.74 to 0.39 over 4 ticks. FI = 0.61. One temporal anchor active; no semantic anchors."

Stage 1 — Intake: Substrate typed as Signal-dominant (coherence was 0.74 at start; intent-bearing content present). Intake Manifest:
substrate_type=Signal, entropy_index=0.29, flux_index=0.61,
timestamp=tick_04.

Stage 2 — Parse: Parse tree generated at mid-depth (FI=0.61 triggers mid-depth parse to capture relational shifts). 8 parse units identified. Units 5–8 show a frame transition: temporal reasoning markers absent in units 6–8; causal

connectives appear with no prior introduction.

Stage 3 — Evaluate: Units 1-5 evaluated: semantic_deviation avg 0.14, flux_weight=0.61, anchor_presence=0.4 → DS avg = $(0.14 \times 0.61) / 0.4 = 0.21$. Clean. Units 6-8 evaluated: semantic_deviation avg 0.58, flux_weight=0.61, anchor_presence=0.1 (temporal anchor does not stabilize semantic frame) → DS avg = $(0.58 \times 0.61) / 0.1 = 3.54$. Major drift candidates flagged.

Stage 4 — Resolve: DS=3.54 far exceeds drift_threshold=0.38. NDP: this is not noise — semantic content is present and coherent within the causal frame. Therefore Suppress is incorrect. Historical DS variance does not indicate threshold miscalibration. Decision: **Escalate** to Triadic Loop.

Stage 5 — Emit: Emit mode=logged. Output: corrected substrate object (units 1-5 clean; units 6-8 flagged pending triadic resolution). Drift log entry: {tick:04, units:[6,7,8], DS:3.54, pattern:sudden, dimensions:[D2], resolution:escalated, outcome:pending}. No pulse signal (tick 4 ≠ multiple of 8).

Practice Task 2.1 — Resolve Stage Decision Tree Application

A parse tree fragment contains 6 parse units. Units 3 and 5 score above drift_threshold. Apply the Resolve stage decision tree to each flagged unit given the context below.

Unit 3: DS = 0.41. The current substrate stream has a long history of producing DS values between 0.38-0.44 due to high flux environment (FI=0.72). Historical DS variance is 0.06. Anchor presence = 0.6.

Decision for Unit 3:

Justification:

Unit 5: DS = 0.89. NDP confirms structured noise signature: mechanical repetition interval of 2 ticks, zero semantic coherence across adjacent units. No intent vector detectable.

Decision for Unit 5:

Justification:

Practice Task 2.2 — Emit Stage Output Description

Based on the Worked Example 2 above, describe what the complete Emit stage output would look like. Include the three required output components and write out the drift_log entry fields in structured format.

Write the three Emit outputs and drift_log entry here:

🔍 Clarity Pulse Checkpoint — Session 2

Q1. What four fields does the Intake Manifest record?

Q2. Define a parse unit in your own words.

Q3. Write the Drift Score formula from memory.

Q4. Name the three resolution strategies in the Resolve stage.

Q5. What three outputs does the Emit stage always produce?

Session 3 of 8 | R5 Pillars: R2, R3 | Standalone: Yes

Drift Detection: Pattern Recognition & Threshold Calibration

Learning Objective

Recognize the three primary drift pattern types (Gradual, Sudden, Recurring) and calibrate the `drift_threshold` using the sigmoid calibration curve for a given substrate environment.

Concept Exposition — Drift Pattern Types

Pattern 1: Gradual Drift

Gradual drift is the most insidious pattern. It is characterized by slow, monotonic deviation — each individual DS value is sub-threshold, but the substrate is steadily moving away from its baseline. Because no single evaluation tick triggers a drift flag, gradual drift can persist for many ticks before it is detected, by which point the cumulative deviation may be severe.

The detection mechanism for gradual drift is **Cumulative Drift Scoring (CDS)**: the sum of DS values over a rolling evaluation window. Gradual drift is triggered when:

Cumulative Drift Score — Gradual Drift Trigger Condition

$$\text{CDS} = \Sigma(\text{DS values over window}) > \text{threshold} \times \text{window_length}$$

For example, with $\text{drift_threshold} = 0.38$ and $\text{window_length} = 4$, gradual drift triggers when $\text{CDS} > 1.52$. Individual DS values may all be 0.35 (sub-threshold), but their cumulative sum of 1.40 approaches the trigger at 4 ticks and exceeds it if even one tick sees a slightly elevated DS.

Pattern 2: Sudden Drift

Sudden drift is an abrupt substrate discontinuity: a single DS spike substantially above the drift_threshold . It is the easiest pattern to detect — the DS spike immediately flags as a Drift Candidate — but paradoxically the hardest to resolve. Sudden drift events often occur in low-anchor environments, precisely because anchors are what would have slowed or prevented the sudden discontinuity. With few anchors present, the resolution options are limited: Escalate is almost always required.

Sudden drift events are identified by: $\text{DS} \geq 2 \times \text{drift_threshold}$ in a single tick, with no preceding elevated DS values (CDS was below trigger for previous window). The absence of a gradual run-up distinguishes sudden from gradual drift.

Pattern 3: Recurring Drift

Recurring drift is the pattern that indicates systemic failure. It manifests as periodic return to a drifting state after what appeared to be successful resolution — the drift_log shows oscillating DS values: high $\rightarrow$ resolved $\rightarrow$ high $\rightarrow$ resolved $\rightarrow$ high. This oscillation confirms that the resolution strategy applied was addressing symptoms, not root cause. The substrate has an underlying structural instability that each resolution temporarily masks but does not eliminate.

Recurring drift requires one of two responses: **permanent recalibration** (if the recurring pattern is a calibration artifact — the threshold is simply wrong for this environment) or **substrate rearchitected** (if the recurrence reflects a genuine structural instability that cannot be corrected through threshold adjustment alone).

Threshold Calibration: The Sigmoid Calibration Curve

The drift_threshold (default 0.38) is not a fixed line. In the RTT_DRIFT_ENGINE, the effective threshold is computed through a sigmoid calibration curve that adapts based on the historical DS variance of the current substrate stream. This prevents both over-sensitivity (too many false positives in inherently high-DS environments) and under-sensitivity (missed real drift in low-variance environments where even a small DS spike is meaningful).

Sigmoid Calibration Curve Formula

$$T(x) = 1 / (1 + e^{(-k(x - x_0))})$$

Where **k** is the steepness parameter (how sharply the sigmoid transitions — higher k = more sensitive threshold; lower k = more gradual transition) and **x₀** is the inflection point (the DS value at which the sigmoid reaches 0.50 — effectively the effective threshold center). Practitioners adjust k to control sensitivity: a high-noise environment might use k=1.2 to reduce sensitivity, while a precision evaluation environment might use k=3.5 to create a sharp decision boundary.

Calibration Worksheet

Environment Type	Observed DS Variance	Recommended k	Resulting Threshold Range	Notes
Low-noise, high-precision	< 0.05	3.5 - 4.0	0.33 - 0.43	Sharp boundary; minimal false positives/negatives
Moderate-noise, standard	0.05 - 0.12	2.0 - 2.5	0.28 - 0.48	Default configuration; balanced sensitivity
High-noise, high-flux	0.12 - 0.20	1.2 - 1.8	0.22 - 0.54	Broad boundary; prioritizes

Environment Type	Observed DS Variance	Recommended k	Resulting Threshold Range	Notes
				recall over precision
Practitioner fill-in				
Practitioner fill-in				

Worked Example 3 — Gradual Drift Detection via CDS

DS time series: [0.22, 0.29, 0.31, 0.35, 0.40, 0.38, 0.41]

$\text{drift_threshold} = 0.38$, $\text{window_length} = 4$, $\text{CDS trigger} = 0.38 \times 4 = 1.52$

Window 1 (ticks 1-4): $\text{CDS} = 0.22 + 0.29 + 0.31 + 0.35 = 1.17$. No trigger ($1.17 < 1.52$).

Window 2 (ticks 2-5): $\text{CDS} = 0.29 + 0.31 + 0.35 + 0.40 = 1.35$. No trigger ($1.35 < 1.52$). Note: tick 5 individually exceeds threshold ($0.40 > 0.38$) and would flag as a Sudden Drift candidate — but the cumulative window has not yet triggered, so this might be an isolated anomaly rather than gradual drift.

Window 3 (ticks 3-6): $\text{CDS} = 0.31 + 0.35 + 0.40 + 0.38 = 1.44$. No trigger ($1.44 < 1.52$). Gradual drift pattern is building.

Window 4 (ticks 4-7): $\text{CDS} = 0.35 + 0.40 + 0.38 + 0.41 = 1.54$. **TRIGGER** ($1.54 > 1.52$). Gradual drift confirmed at tick 7.

Interpretation: Gradual drift triggers at tick 7. No single tick before tick 5 would have flagged individually, demonstrating why CDS is essential for this pattern type. The practitioner should now run NDP to determine if this is structural noise or genuine semantic drift, then apply the appropriate resolution.

Practice Task 3.1 — Drift Pattern Classification

Classify each scenario as **Gradual**, **Sudden**, or **Recurring** drift. Justify your classification.

Scenario A: DS series over 10 ticks: [0.15, 0.17, 0.14, 0.82, 0.11, 0.16, 0.13, 0.18, 0.15, 0.14]. No preceding elevated CDS window.

Classification:

Justification:

Scenario B: drift_log shows DS history over 40 ticks: Low (0.15–0.22) for ticks 1–8, resolved escalation at tick 9, low again 10–18, resolved escalation at tick 20, low 21–30, escalation at tick 32. Pattern repeating every ~12 ticks.

Classification:

Justification:

Scenario C: DS series: [0.20, 0.23, 0.26, 0.28, 0.30, 0.32, 0.35, 0.36, 0.38, 0.40]. Monotonically increasing over 10 ticks with no single spike. Window 4 = ticks 7–10: CDS = 1.49. Window 5 = ticks 7–10 extended by one tick if next DS ≥ 0.04 would trigger.

Classification:

Justification:

Practice Task 3.2 — Sigmoid Threshold Calculation

Given calibration parameters: $k = 2.1$, $x_0 = 0.38$. Compute $T(x)$ for $x = 0.30$, $x = 0.38$, and $x = 0.50$. Round to 4 decimal places.

$$T(0.30) = 1 / (1 + e^{(-2.1(0.30 - 0.38))}) =$$

$$T(0.38) = 1 / (1 + e^{(-2.1(0.38 - 0.38))}) =$$

$$T(0.50) = 1 / (1 + e^{(-2.1(0.50 - 0.38))}) =$$

What do these three values tell you about the effective threshold curve at $k=2.1$?

◻ Clarity Pulse Checkpoint — Session 3

Q1. What distinguishes Gradual drift from Sudden drift?

Q2. What is Cumulative Drift Scoring (CDS) and what is its trigger condition?

Q3. What drift pattern indicates that the resolution strategy failed to address root cause?

Q4. What does the sigmoid calibration curve adapt to, and what does the parameter k control?

Dimensional Curriculum: Navigating the 5 Dimensions of Drift

Learning Objective

Apply the five dimensional axes (D1-D5) to contextualize a drift signal, identify the primary drifting dimension, and determine the appropriate dimensional response sequence.

Concept Exposition — The Five Dimensional Axes

D1 — Temporal Dimension: Drift in Time

The temporal dimension governs drift as a phenomenon in time. Temporal drift means the substrate's signal is misaligned with the expected time-anchored sequence — events appear out of order, timestamps are anomalous, or temporal references within the substrate contradict the evaluation timeline. Temporal drift is often the earliest detectable form of drift because it is structural: a substrate's temporal logic must be consistent before its semantic content can be evaluated reliably.

Key Indicators: timestamp anomalies in the Intake Manifest; parse units appearing out of expected sequence; temporal anchor collapse (a previously active temporal anchor becomes inconsistent with new substrate); out-of-order causal chains in the parse tree.

Primary Response: Reassert temporal anchor. If no temporal anchor is available, assign one using the most recent stable timestamp as the reference point. Re-sequence the parse tree against the reasserted anchor before proceeding with Evaluate.

D2 — Semantic Dimension: Drift in Meaning

The semantic dimension governs drift as meaning-deviation. Semantic drift occurs when parse units lose coherent meaning relative to the established intent vector — the substrate still has content, but that content is no longer aligned with the purpose or direction established at the start of the evaluation. Semantic drift is the most common form encountered in practice.

Key Indicators: semantic coherence score < 0.45 ; frame-shift events (the conceptual frame of reference changes without transition signaling); referential ambiguity spikes (pronouns, demonstratives, or conceptual references that cannot be resolved to a clear antecedent).

Primary Response: Re-establish semantic anchor using the intent vector baseline from the Intake Manifest. Apply deep parse (depth 3) to the drifting parse units to extract the new intent vector they are expressing. If the new vector is legitimate (intentional frame shift), update the baseline. If it is drift, escalate.

D3 — Relational Dimension: Drift in Connections

The relational dimension governs drift in the relationships between parse units. Relational drift means the links between concepts, agents, or propositions have become unstable — not because individual units are semantically deviant, but because the map of how they connect to each other is degrading. A parse tree with relational drift may still contain coherent individual units that simply no longer connect meaningfully to each other.

Key Indicators: relational entropy > 0.55 (measure of disorder in the parse tree's relational links); orphaned parse units (units with no relational links to other units — they exist in isolation within the parse tree); circular relational patterns (unit A links to B, B links to C, C links to A, forming a loop with no external connections).

Primary Response: Rebuild relational map. Identify all orphaned units and attempt to establish relational links through mid or deep parse. Assign relational anchors to high-connectivity nodes (parse units with the most relational links) to prevent further link degradation.

D4 — Intentional Dimension: Drift in Purpose

The intentional dimension governs drift in the intent vector itself — the most severe and fundamental form of drift. Intentional drift does not mean the substrate's surface content has changed; it means the substrate's underlying purpose has shifted. D4 drift implies that what the substrate is trying to accomplish has changed, which invalidates all prior dimensional analyses that were calibrated to the original intent.

Key Indicators: intent vector cosine similarity < 0.30 against the baseline intent vector established at Intake (a cosine similarity below 0.30 indicates the current intent vector is nearly orthogonal to the baseline — a fundamentally different direction); goal-frame collapse (the substrate's action-goal structure is internally incoherent); action-outcome misalignment (the actions described or implied by the substrate do not produce the outcomes the substrate's stated goals would require).

Primary Response: Full triadic loop escalation. D4 drift cannot be resolved through threshold recalibration or suppression — the intent vector must be re-established through the full Perceive-Process-Project cycle. In severe cases, substrate rearchitecting may be required before triadic loop processing can proceed effectively.

D5 — Integrative Dimension: Meta-Level Drift

The integrative dimension operates at the meta-level — it governs the system's ability to integrate signals across D1–D4 simultaneously. Integrative drift is not drift in any single dimension; it is drift in the system's capacity to hold all dimensions in coherent relationship. It manifests as cross-dimensional interference: corrective actions in one dimension that worsen drift in another, creating a situation where the system cannot improve overall without first resolving the integration failure.

Key Indicators: corrective actions increasing DS scores across multiple dimensions simultaneously (the primary indicator — fixing one thing breaks another); D1 corrective actions worsening D2 coherence, or D3 repair destabilizing D4 intent vector; DS scores across multiple dimensions all above threshold simultaneously.

Primary Response: Pause all corrective actions immediately. Any further targeted dimensional corrections will worsen integrative drift. Enter Clarity Pulse mode:

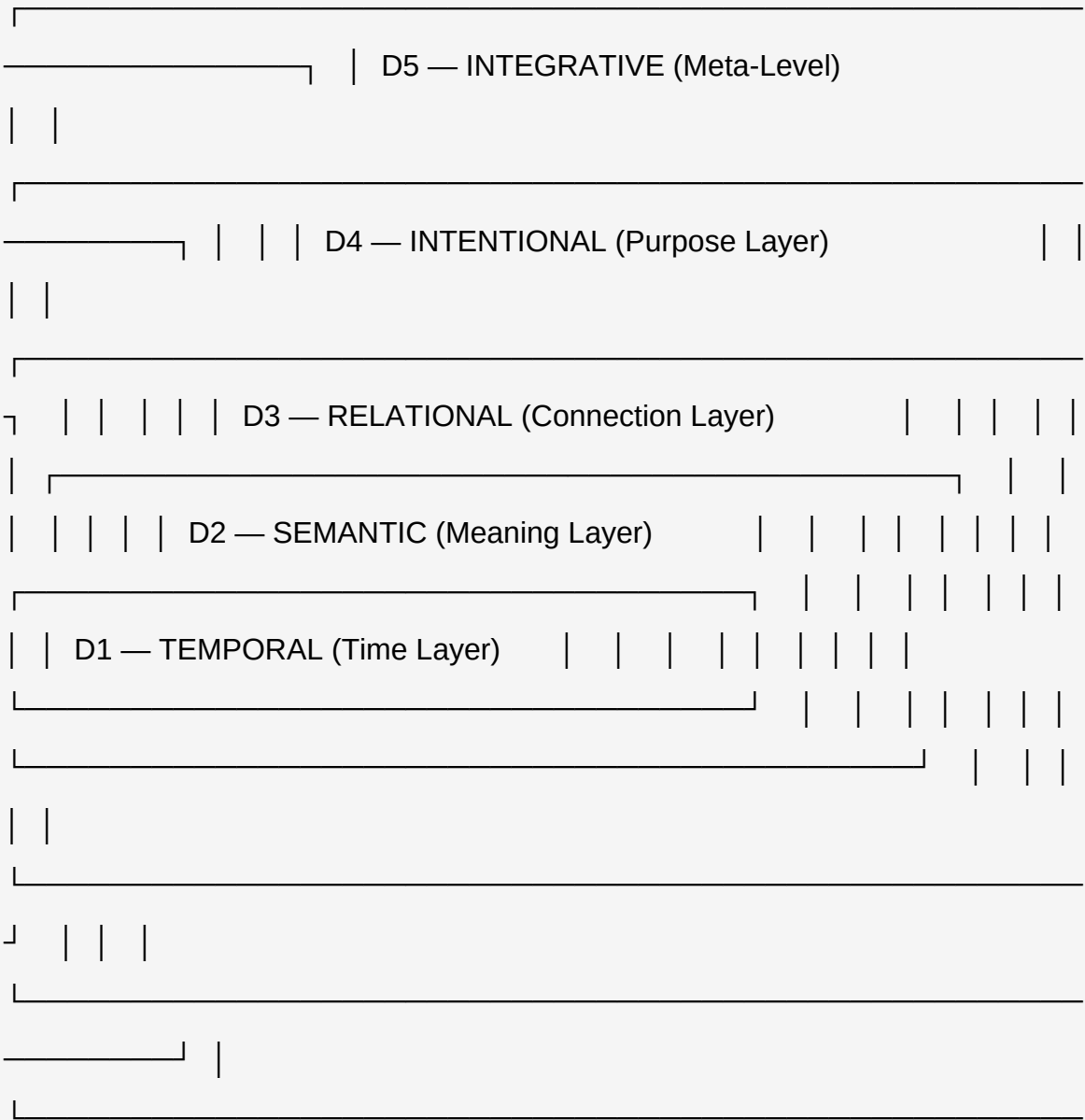
trigger a full CPC regardless of tick position. Run a full dimensional audit across all five axes before re-engaging any resolution strategy. The dimensional audit must identify which dimension is the root instability (usually the dimension whose repair consistently worsens the others).

Dimensional Scoring Rubric

Dim.	Key Indicator	Threshold	Primary Response	Escalation Trigger
D1 Temporal	Timestamp anomalies; out-of-sequence parse units; temporal anchor collapse	Any temporal anchor collapse; sequence anomaly in >2 consecutive units	Reassert temporal anchor; re-sequence parse tree	Temporal anchor collapse with no available reference timestamp
D2 Semantic	Semantic coherence score; frame-shift events; referential ambiguity	Coherence < 0.45; frame shift without transition signal	Re-establish semantic anchor; apply deep parse	Coherence < 0.30; multiple simultaneous frame shifts
D3 Relational	Relational entropy; orphaned parse units; circular patterns	Relational entropy > 0.55; any orphaned parse unit	Rebuild relational map; assign relational anchors	Relational entropy > 0.75; majority of units orphaned
D4 Intentional	Intent vector cosine similarity; goal-frame collapse; action-outcome misalignment	Cosine similarity < 0.30 against baseline	Full triadic loop escalation	Cosine similarity < 0.10; complete goal-frame collapse
D5 Integrative	Corrective actions worsening DS across multiple dimensions simultaneously	Any cross-dimensional worsening from corrections	Pause corrections; enter Clarity Pulse mode; full dimensional audit	Three or more dimensions simultaneously above threshold with interdependent corrections

Dimensional Interaction Diagram

DIMENSIONAL ARCHITECTURE — DRIFT PROPAGATION MAP



DRIFT PROPAGATION DIRECTION (typical): D1 anomaly → may destabilize D2 (temporal frame collapse → semantic coherence loss) D2 drift → may propagate to D3 (meaning loss → relational link degradation) D3 drift → may propagate to D4 (relational collapse → intent vector corruption) D4 drift → may trigger D5 (intent shift causes cross-dimensional interference) IMPORTANT: Propagation is not inevitable — anchors at each layer contain drift within that dimension. Only in low-anchor environments does

drift consistently propagate upward through the stack.

Worked Example 4 — Dimensional Triage

Observed indicators: Timestamp anomalies present (D1); semantic coherence = 0.41 (D2 flag at <0.45); 3 orphaned parse units (D3 flag); intent cosine similarity = 0.44 against baseline (D4: not yet flagged at >0.30, but approaching); corrective actions applied to D1 (temporal re-sequencing) are causing semantic coherence to drop further to 0.38 (worsening D2 — D5 signal).

Step 1 — Dimensional Triage: D1 flag (timestamp anomalies), D2 flag (coherence 0.41), D3 flag (orphaned units), D5 emerging signal (correction worsening another dimension). D4 at 0.44 is approaching but not yet triggered. Priority assessment: D5 emerging takes precedence — the cross-dimensional interference means targeted corrections are making things worse.

Step 2 — Primary Dimension: D5 Integrative drift is the primary concern, not because it is the most severe in isolation, but because it prevents resolution of the lower-dimensional flags. Any further D1 or D2 corrections will continue to interfere with each other.

Step 3 — Resolution Sequence: (1) Immediately pause all corrective actions. (2) Trigger CPC regardless of tick position. (3) Run full dimensional audit across D1-D4 to identify which dimension's instability is driving the cross-dimensional interference. (4) Identify that D3 orphaned units are the root cause — they are creating relational voids that both the D1 re-sequencing and D2 semantic anchoring are fighting to fill, producing contradictory corrections. (5) Resolve D3 first: rebuild relational map, link orphaned units. (6) Re-evaluate D1 and D2 after D3 stabilization. (7) Re-assess D4 cosine similarity once lower layers are stable.

Practice Task 4.1 — Primary Dimension Identification

For each scenario, identify the **primary drifting dimension** and the **correct first response**.

Scenario 1: Parse units 4–6 appear before parse unit 3 in the sequence. Temporal anchor assigned at tick 1 is now inconsistent with the parse tree ordering.

Primary Dimension: First Response:

Scenario 2: Semantic coherence = 0.39. Four consecutive frame-shift events detected without transition signaling. Intent vector still intact at cosine similarity 0.67.

Primary Dimension: First Response:

Scenario 3: Relational entropy = 0.68. Six of eleven parse units are orphaned. No timestamp anomalies. Semantic coherence = 0.62 (healthy).

Primary Dimension: First Response:

Scenario 4: Intent vector cosine similarity = 0.21. Goal-frame collapse confirmed. Action-outcome misalignment in 3 of 4 goal structures.

Primary Dimension: First Response:

Scenario 5: DS increasing across D2 and D3 simultaneously despite separate corrective actions having been applied to each. Each fix worsens the other dimension.

Primary Dimension: First Response:

Practice Task 4.2 — Simultaneous D2 and D4 Drift: Risk Analysis

A drift event shows indicators in D2 (semantic coherence = 0.38, multiple frame shifts) and D4 (intent cosine similarity = 0.27, goal-frame partially collapsed) simultaneously. Explain the risk this poses and describe what dimensional response sequencing is required.

Risk analysis (write here):

Response sequence (step by step):

Step 1:

Step 2:

Step 3:

Step 4:

📍 Clarity Pulse Checkpoint — Session 4

Q1. In one sentence, define D4 intentional drift.

Q2. What threshold value distinguishes healthy from drifting relational entropy (D3)?

Q3. What is the primary indicator of D5 integrative drift?

Q4. Why must you pause all corrective actions during D5 drift rather than intensifying them?

Triadic Learning Loops: Perceive → Process → Project

Learning Objective

Execute a complete triadic learning loop cycle — including Perception Report generation, strategy ranking via RIS, and monitored projection — in response to an escalated drift event.

Concept Exposition — The Triadic Loop

The triadic loop is the Drift Engine's core corrective intelligence mechanism. It activates when the Resolve stage escalates a Drift Candidate — when neither Recalibrate nor Suppress is appropriate and the drift requires deeper analysis and targeted intervention. The loop is called "triadic" because it has exactly three phases that must execute in strict sequence: Perceive, Process, and Project. Skipping or compressing phases is the most common source of loop failure in practice.

Phase 1: PERCEIVE

The Perceive phase is about gathering complete situational awareness without acting. The practitioner's (or system's) entire focus in this phase is on building an accurate, unbiased picture of the drift event. No resolution hypotheses should be formed; no strategies should be mentally committed to. Premature resolution degrades loop integrity because it introduces confirmation bias into the Process

phase — the practitioner begins looking for evidence that supports their pre-formed strategy rather than genuinely evaluating all options.

Perceive Phase Actions:

- Assemble the full drift context: substrate type from Intake Manifest, current DS score, active dimensional flags (D1-D5), drift pattern type (Gradual/Sudden/Recurring), and anchor inventory (which anchors are present, absent, or degraded).
- Generate the Perception Report — the structured output of the Perceive phase.
- Default duration: 3 evaluation cycles. This can be configured in the triadic_loop object.

Perception Report Fields:

Field	Description
event_id	Unique identifier for this drift event
substrate_type	From Intake Manifest
current_DS	DS score at time of escalation
dimensional_flags	Array of active dimension flags (e.g., [D2, D3])
drift_pattern	Gradual / Sudden / Recurring
anchor_inventory	Active anchors by type and stability rating
flux_index	Current FI at time of escalation
perceive_duration	Number of cycles spent in Perceive phase

Phase 2: PROCESS

The Process phase is the analysis and strategy-generation phase. The Perception Report is the input. The output is a ranked set of 2-3 candidate resolution strategies, each with a projected DS impact and a computed Resolution Impact Score (RIS). The key discipline of the Process phase is restraint: **never commit to a strategy in the Process phase**. The output is a ranked strategy set, not a decision. The decision is made in Project.

Resolution Impact Score (RIS) Formula

$$\text{RIS} = (\text{projected_DS_reduction} \times \text{anchor_stability}) / \text{resource_cost}$$

Where **projected_DS_reduction** is the estimated decrease in DS if this strategy is applied (range 0-1), **anchor_stability** is the estimated stability of the anchor state after strategy application (range 0-1; a strategy that depletes or destabilizes anchors scores lower), and **resource_cost** is the relative operational cost of executing the strategy (range 0.1-1.0; lower cost strategies score higher).

Phase 3: PROJECT

The Project phase is the action and verification phase. The top-ranked strategy from Process is selected and projected into the substrate. Projection means applying the strategy through the Resolve stage — either recalibrating the threshold, suppressing identified noise, or executing the specific triadic correction developed in Process.

Post-Projection Monitoring: For 5 ticks following projection, DS values are monitored. Success criterion: DS must decrease by >0.10 within the monitoring window. If this criterion is not met, the projection is classified as failed and loop re-entry is triggered.

Loop Re-entry Conditions

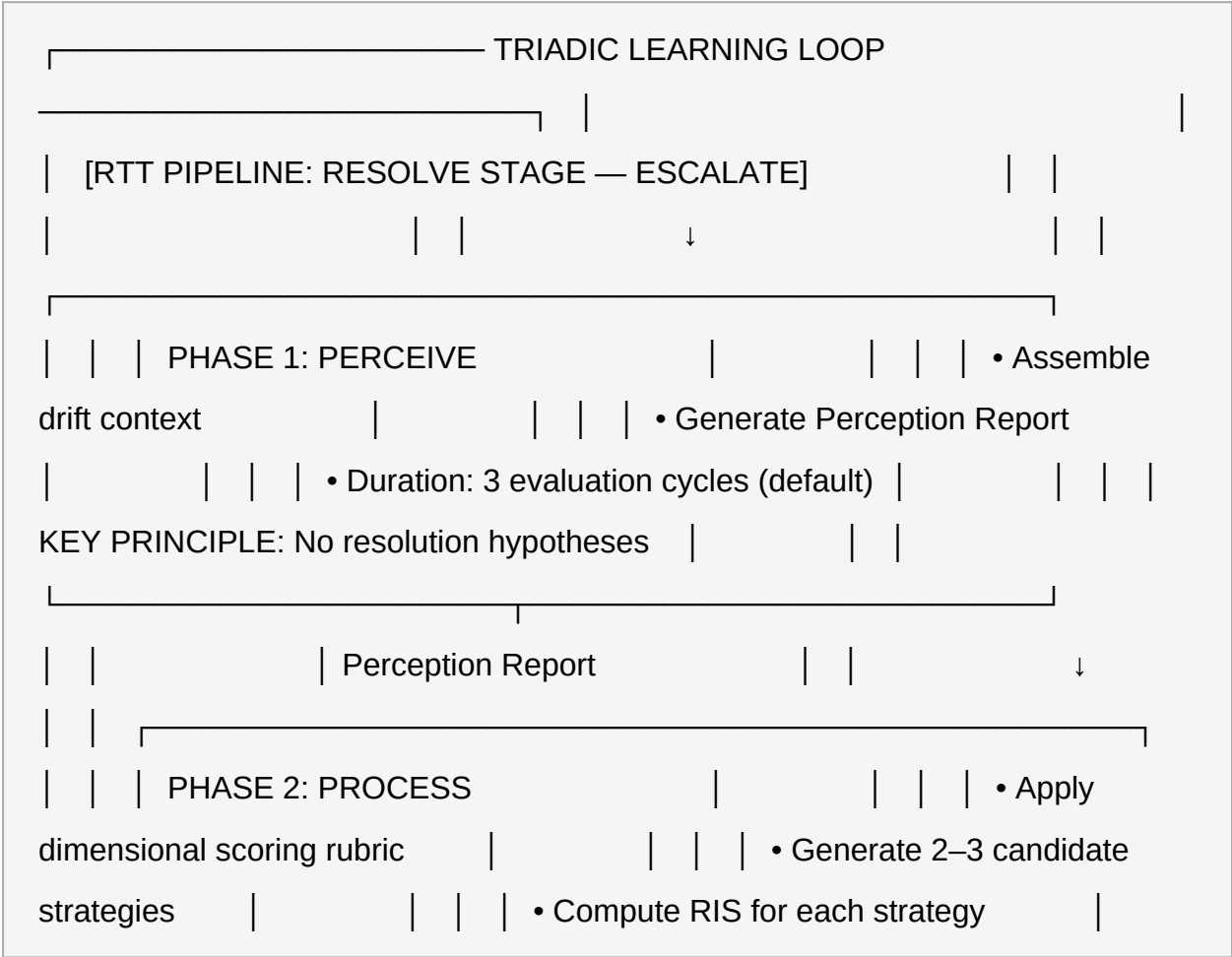
The following four conditions force mandatory loop re-entry:

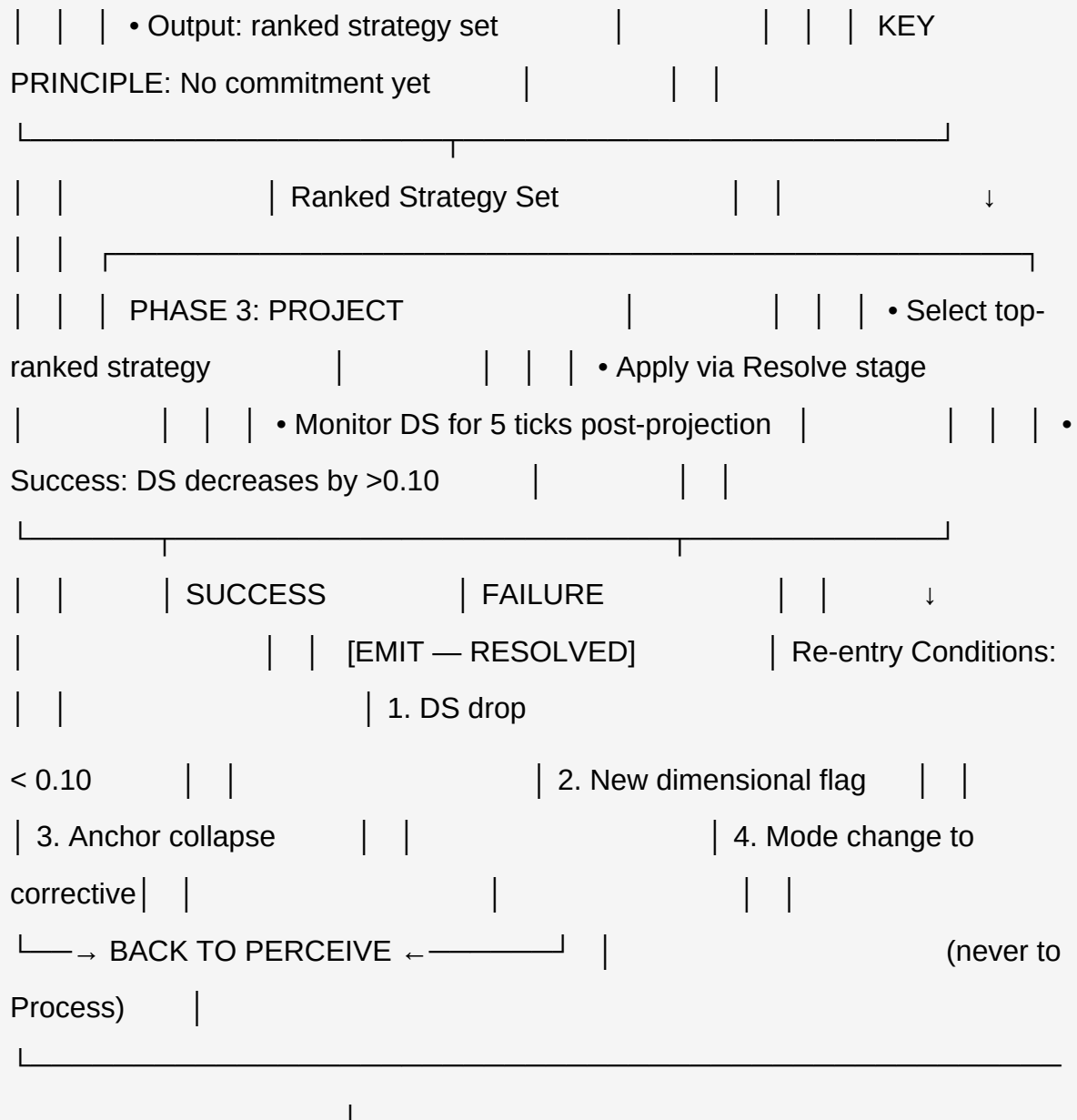
15. **DS fails to decrease by >0.10 within 5 ticks post-projection.** The strategy did not achieve its projected impact.
16. **A new dimensional flag emerges during the projection phase.** The projection disturbed the substrate in an unanticipated way, creating new drift dimensions that were not present in the Perception Report.
17. **Anchor collapse occurs during projection.** A previously stable anchor failed during the corrective intervention, destabilizing the substrate further.

18. **evaluator_mode changes to corrective mid-loop.** A mode change indicates external escalation; the loop must re-perceive with updated authority context.

Critical Rule: When loop re-entry is triggered, re-entry ALWAYS begins at **Perceive**, not at Process. The failed projection has changed the substrate state. The Perception Report from the prior cycle is now stale — it describes a substrate that no longer exists. Fresh perception is required before any new strategy generation.

Triadic Loop Diagram





Worked Example 5 — Full Triadic Loop: Semantic Drift, High Flux

Escalated event: D2 semantic drift, DS=0.52, no temporal anchors present, Flux Index = 0.71. Drift pattern: Sudden (single spike from baseline DS of 0.19).

PERCEIVE Phase: Assemble context. Event ID: EVT_0052. Substrate type: Signal. Current DS: 0.52. Dimensional flags: [D2]. Drift pattern: Sudden. Anchor

inventory: {temporal: NONE, semantic: NONE, relational: 1 (stability: 0.6)}. FI: 0.71. Perceive duration: 3 cycles. *Perception Report generated.* No hypotheses formed in this phase. Note: absence of both temporal and semantic anchors with FI=0.71 is a high-risk configuration.

PROCESS Phase: Apply dimensional scoring rubric: D2 primary (semantic coherence likely below 0.45 given DS=0.52). No D1 flag (despite no temporal anchor — temporal sequence appears intact). No D3 flag (1 relational anchor active). Generate strategies:

Strategy A — Semantic Anchor Assignment: Assign a semantic anchor using the intent vector from the last clean parse unit (tick prior to escalation).

projected_DS_reduction = 0.25 (moderate — anchor will suppress some semantic deviation). anchor_stability = 0.7 (new anchor, not yet tested by flux). resource_cost = 0.3. $RIS = (0.25 \times 0.7) / 0.3 = 0.583$.

Strategy B — Threshold Recalibrate + Temporal Anchor: Recalibrate k upward (reduce sensitivity) and assign temporal anchor to stabilize against FI=0.71.

projected_DS_reduction = 0.15 (lower — recalibration masks rather than corrects). anchor_stability = 0.8. resource_cost = 0.5. $RIS = (0.15 \times 0.8) / 0.5 = 0.240$.

Ranking: Strategy A (RIS=0.583) > Strategy B (RIS=0.240). Strategy A selected.

PROJECT Phase: Apply Strategy A. Semantic anchor assigned at tick of escalation using prior clean parse unit intent vector. Monitor DS for 5 ticks. Results: tick+1: DS=0.44, tick+2: DS=0.39, tick+3: DS=0.33, tick+4: DS=0.29, tick+5: DS=0.24. Total drop: $0.52 - 0.24 = 0.28 > 0.10$. **Success.** Loop exits. Emit stage generates resolved output with drift_log entry noting successful triadic resolution via Strategy A.

Practice Task 5.1 — RIS Computation from Perception Report

A Perception Report has been generated for an escalated drift event. Using the data provided, generate two candidate resolution strategies and compute RIS for each.

Perception Report: Event ID: EVT_0108. Substrate: Signal. DS: 0.67. Dimensional flags: [D3, D1]. Drift pattern: Gradual. Anchor inventory: {temporal: 2 (stability: 0.8, 0.5), semantic: 1 (stability: 0.4), relational: NONE}. FI: 0.49.

Strategy A (your design): Description:

projected_DS_reduction: ____ anchor_stability: ____ resource_cost: ____

$RIS = (\text{ } \times \text{ }) / \text{ } = \text{ }$

Strategy B (your design): Description:

projected_DS_reduction: ____ anchor_stability: ____ resource_cost: ____

$RIS = (\text{ } \times \text{ }) / \text{ } = \text{ }$

Recommended Strategy: Justification:

Practice Task 5.2 — Failed Projection Re-entry Procedure

A projection was applied. Post-projection DS monitoring over 5 ticks shows: [0.65, 0.63, 0.64, 0.62, 0.63]. The starting DS was 0.67. DS has dropped only

0.04 — below the 0.10 success threshold.

Answer the following:

- (a) Is loop re-entry triggered? Yes / No (circle one) Justification:
- (b) At which phase does re-entry begin?
- (c) Why cannot re-entry begin at the Process phase?
- (d) What is the first action in the new Perceive phase?

🏠 Clarity Pulse Checkpoint — Session 5

Q1. What triggers triadic loop activation?

Q2. What is the key principle of the Perceive phase, and why does violating it degrade loop integrity?

Q3. Write the RIS formula from memory.

Q4. Under what condition must re-entry start at Perceive rather than Process?

Q5. How many ticks is the default post-projection monitoring window?

Session 6 of 8 | R5 Pillar: R4 | Standalone: Yes

Clarity Pulse Checkpoints: Real-Time Calibration

Tools

Learning Objective

Deploy and interpret Clarity Pulse Checkpoints (CPCs) within an active drift evaluation environment, including filling all five CPC blocks and rendering a correct Pulse Decision.

Concept Exposition — What Are CPCs?

Clarity Pulse Checkpoints are the Drift Engine's built-in metacognitive layer. They serve as regular structured pauses in the evaluation flow, operating on the `pulse_interval` cadence (default: every 8 ticks). CPCs prevent a dangerous failure mode called **evaluator drift** — the gradual loss of cognitive alignment between the human practitioner and the engine's operational state. Even skilled evaluators can unconsciously drift away from accurate assessment when immersed in a complex, rapidly-changing substrate environment. CPCs are the structural intervention that catches and corrects this drift before it contaminates the evaluation.

CPCs serve three primary functions:

19. **State Verification:** Confirm that the engine's current operational state (DS values, active dimensions, anchor inventory, triadic loop status) matches expected parameters. State Verification catches silent failures — states that are wrong but have not yet triggered an alert.
20. **Calibration Audit:** Check whether the `drift_threshold` and `calibration_curve` parameters remain appropriate for the current substrate environment. As a substrate evolves, the environment characteristics may shift enough that the original calibration is no longer optimal.
21. **Practitioner Alignment:** For human evaluators, CPCs provide the structured moment to step back from the detail level and verify that their

high-level understanding of the drift situation remains accurate. Alignment questions are tailored to the specific drift context active at the time of the CPC.

CPC Anatomy — Five Blocks

Block	Name	Contents	Mandatory?
1	Pulse Header	tick_count, current_DS, active_dimensions, triadic_loop_status	Yes
2	State Verification Block	3 binary (pass/fail) checks against expected parameters	Yes
3	Calibration Audit Block	Current k, x ₀ , threshold range, recommendation flag (yes/no)	Yes
4	Practitioner Alignment Block	3–5 short-answer questions tailored to current drift context	Yes
5	Pulse Decision	Continue / Adjust / Interrupt — derived from blocks 2 and 3	Yes

Pulse Decision Logic

PULSE DECISION DERIVATION: State Verification results: All 3 checks PASS AND Calibration Audit: NO recommendation flag → CONTINUE (no action required; evaluation proceeds) 1–2 checks FAIL OR Calibration Audit flags a recommendation → ADJUST (targeted parameter adjustment; document changes) 3 checks FAIL OR triadic_loop re-entries > 2 → INTERRUPT (halt evaluation; escalate; full audit required)

Five Principles of CPC Design

22. **Specificity:** Every question in the Practitioner Alignment Block must be tied to the specific drift context currently active. Generic questions that could apply to any evaluation do not provide alignment value.
23. **Brevity:** The entire CPC must be completable in under 2 minutes. If a CPC requires more than 2 minutes, it has become an evaluation event in itself — defeating its purpose as a calibration pause.
24. **Sequencing:** Always execute blocks in order: Pulse Header → State Verification → Calibration Audit → Practitioner Alignment → Pulse Decision. Never skip to Pulse Decision without completing the prior blocks.
25. **Non-redundancy:** No CPC question should repeat a question from a prior CPC in the same evaluation session. Each CPC should examine a different aspect of the current state.
26. **Actionability:** Every Practitioner Alignment answer must map to a specific engine parameter or decision. Questions that produce answers with no actionable implication should be replaced.

CPC Template (Fillable)

CLARITY PULSE CHECKPOINT	
BLOCK 1: PULSE HEADER	
tick_count:	
current_DS:	
active_dimensions:	
triadic_loop_status:	
BLOCK 2: STATE VERIFICATION	
Check 1: DS within expected range for current environment?	PASS / FAIL
Check 2: Anchor inventory matches last logged state?	PASS / FAIL
Check 3: Dimensional flags consistent with	PASS / FAIL

CLARITY PULSE CHECKPOINT	
prior assessment?	
BLOCK 3: CALIBRATION AUDIT	
Current k:	
Current x_0 :	
Current threshold range:	
Recommendation flag:	YES / NO — If YES:
BLOCK 4: PRACTITIONER ALIGNMENT	
Q1 (context-specific):	
Q2 (context-specific):	
Q3 (context-specific):	
Q4 (optional):	
BLOCK 5: PULSE DECISION	
CONTINUE ADJUST INTERRUPT	
If ADJUST — action required:	
If INTERRUPT — escalation path:	

Worked Example 6 — Complete CPC at Tick 16

Context: Tick 16 of an active evaluation. $\text{pulse_interval} = 8$, so CPC triggers. Current DS = 0.44 (down from 0.52 at tick 8). Triadic loop ran once: projected at tick 10, succeeded at tick 12 (DS dropped 0.16 in 5 ticks). Active dimensions: [D2]. No new dimensional flags since tick 12. Calibration: $k=2.0$, $x_0=0.38$.

Block 1 — Pulse Header: $\text{tick_count}=16$, $\text{current_DS}=0.44$, $\text{active_dimensions}=[D2]$, $\text{triadic_loop_status}=\text{completed_once}$ (last projection: success at tick 12).

Block 2 — State Verification: Check 1: DS=0.44 is above threshold (0.38) — is this expected? Yes, D2 drift was escalated and resolved; residual DS of 0.44 during recovery period is within expected range given FI=0.51 (moderate). PASS. Check 2: Anchor inventory at tick 16: semantic anchor (assigned tick 9,

stability 0.7). Matches logged state. PASS. Check 3: D2 flag active; no D3 or D4 flags. Consistent with prior assessment. PASS.

Block 3 — Calibration Audit: $k=2.0$, $x_0=0.38$, threshold range $\approx 0.28-0.48$. Historical DS variance for this stream: 0.08 (moderate). $k=2.0$ is appropriate for this variance level. Recommendation flag: NO.

Block 4 — Practitioner Alignment: Q1: What was the primary trigger for the triadic loop at tick 8–9? (Expected: sudden D2 semantic drift, $DS=0.52$, no semantic anchor.) Q2: The current DS of 0.44 is still above threshold — should the evaluation escalate again? (Expected: No — DS is declining post-triadic resolution; monitor for further decrease before re-escalating.) Q3: At what tick will the next CPC fire? (Expected: tick 24.)

Block 5 — Pulse Decision: All three State Verification checks PASS. Calibration Audit: NO flag. → **CONTINUE.**

Practice Task 6.1 — Design a CPC for Post-D3 Resolution

D3 relational drift has just been resolved via anchor reassignment at tick 32. The pulse_interval triggers a CPC at tick 32. Design a complete CPC filling in all five blocks. Current state: DS dropped from 0.61 to 0.29 post-resolution. Relational entropy reduced from 0.68 to 0.38. 2 relational anchors now active (stability 0.8 and 0.6). No other dimensional flags. $k=2.0$, $x_0=0.38$.

Block 1 — Pulse Header:

tick_count: current_DS: active_dimensions: triadic_loop_status:

Block 2 — State Verification (3 checks with PASS/FAIL):

Check 1:

Check 2:

Check 3:

Block 3 — Calibration Audit:

k: x₀: Threshold range: Recommendation flag:

Block 4 — Practitioner Alignment (3 context-specific questions):

Q1:

Q2:

Q3:

Block 5 — Pulse Decision: Justification:

Practice Task 6.2 — Interrupt Pulse Decision Response

A CPC Pulse Decision comes back INTERRUPT. What must happen next? Detail the three mandatory steps in sequence.

Step 1:

Step 2:

Step 3:

🏠 Clarity Pulse Checkpoint — Session 6

Q1. What are the three functions of a Clarity Pulse Checkpoint?

Q2. What two conditions trigger an Interrupt Pulse Decision?

Q3. Name the five principles of CPC design.

Q4. What is the correct block sequencing within a CPC?

Session 7 of 8 | R5 Pillar: R5 | Standalone: Yes

Evaluator Exercises: Applied Drift Scoring & Correction

Learning Objective

Act as a full evaluator in `evaluator_mode: active` — scoring drift across all engine layers, applying corrections, and producing properly formatted `drift_log` documentation for all findings.

Evaluator Role Description

In `evaluator_mode: active`, the practitioner acts as a human-in-the-loop evaluator working alongside the engine. The evaluator does not replace the engine's automated processing — they verify it. The active evaluator's responsibilities are:

- Verify Intake Manifest accuracy (substrate type classification, EI, FI)

- Validate parse unit classifications (confirm parse tree is correctly structured)
- Check DS scores against threshold (confirm Evaluate stage calculations)
- Confirm dimensional flags (ensure the correct dimensions are active)
- Review triadic loop strategy selection (validate RIS calculations and strategy ranking)
- Sign off on Emit output (confirm that corrected substrate object and drift_log entry are accurate)

Session 7 Note: This session contains no new conceptual exposition. It is entirely exercise-based and tests cumulative mastery of Sessions 1-6. All formulas, decision trees, and rubrics from prior sessions are available for reference in the appendices. Practitioners should attempt each exercise independently before consulting reference materials.

Exercise 7.1 — Full Pipeline Evaluation

Substrate Description: An input stream arrives at tick 23. Characteristics: the substrate is a complex signal containing a mixture of temporal reasoning (approximately 60% of parse units reference time-ordered events) and emergent causal claims (approximately 40% introduce causal relationships not present in prior ticks). Signal entropy at arrival: 0.31. Over the prior 6 ticks, the signal entropy had been stable at 0.18-0.22. Flux in the most recent 3 ticks: $\Delta\text{signal} = 0.22$, $\Delta\text{time} = 3$ ticks. Anchor inventory: 2 temporal anchors (stability 0.7 and 0.6), 1 semantic anchor (stability 0.55), 0 relational anchors.

(a) Complete the Intake Manifest:

substrate_type: _____

entropy_index: _____

flux_index: _____

timestamp: _____

(b) Drift Candidate Identification: The parse tree contains 9 parse units. Parse units 6, 7, and 8 exhibit the following characteristics: Unit 6: semantic_deviation=0.44, flux_weight=[from your Intake Manifest calculation], anchor_presence=0.3. Unit 7: semantic_deviation=0.51, flux_weight=[same], anchor_presence=0.25. Unit 8: semantic_deviation=0.38, flux_weight=[same], anchor_presence=0.4. Compute DS for each and identify which are Drift Candidates.

DS (Unit 6) = (____ × ____) / ____ = ____ Drift Candidate? YES / NO

DS (Unit 7) = (____ × ____) / ____ = ____ Drift Candidate? YES / NO

DS (Unit 8) = (____ × ____) / ____ = ____ Drift Candidate? YES / NO

(c) Assign Dimensional Flags: Based on the substrate description and parse unit analysis, which dimensional axes (D1-D5) should be flagged? Justify each flag.

Active Dimensions: _____

Justification:

(d) Select Resolution Strategy for each Drift Candidate: Apply the Resolve stage decision tree.

Unit 6 Resolution: _____ *Justification:* _____

Unit 7 Resolution: _____ *Justification:* _____

Unit 8 Resolution: _____ *Justification:* _____

(e) Write the Drift Log Entry: Format as structured key-value pairs.

tick: ____ */ units:* ____ */ DS_values:* ____ */ pattern:* ____ */ dimensions:* ____ */ resolution:* ____ */ outcome:* ____

Exercise 7.2 — Drift Pattern Classification Challenge

Classify each DS time-series as **Gradual**, **Sudden**, **Recurring**, or **No Drift**. For Cases 4 and 7 (marked ambiguous), CDS calculation is required.

drift_threshold=0.38, window_length=4.

Case	DS Time-Series	Classification	Notes / CDS (if required)
1	[0.12, 0.14, 0.11, 0.16, 0.13, 0.15, 0.12, 0.14]		
2	[0.18, 0.21, 0.19, 0.22, 0.91, 0.20, 0.18, 0.19]		
3	[0.22, 0.19, 0.55, 0.21, 0.20, 0.58, 0.22, 0.19, 0.61]		
4 ★	[0.28, 0.30, 0.33, 0.34, 0.36, 0.37, 0.38, 0.40]		Compute CDS windows 1–5
5	[0.15, 0.17, 0.72, 0.68, 0.71, 0.69, 0.70, 0.72]		
6	[0.30, 0.28, 0.55, 0.29, 0.30, 0.52, 0.31, 0.30, 0.54]		
7 ★	[0.25, 0.28, 0.31, 0.35, 0.37, 0.36, 0.38, 0.35]		Compute CDS windows 2–5
8	[0.41, 0.44, 0.43, 0.46, 0.42, 0.45, 0.44, 0.43]		

Exercise 7.3 — Triadic Loop Simulation: D4 Intentional Drift

Pre-written Perception Report: Event ID: EVT_0227. Substrate: Signal. Current DS: 0.74. Dimensional flags: [D4]. Drift pattern: Sudden. Anchor inventory: {temporal: 1 (stability 0.5), semantic: 2 (stability 0.7, 0.6), relational: 1 (stability 0.8)}. FI: 0.38. Intent cosine similarity: 0.22 against baseline. Goal-frame collapse confirmed. perceive_duration: 3 cycles.

(a) Generate two candidate resolution strategies:

Strategy A: _____

Strategy B: _____

(b) Compute RIS for each:

$RIS(A) = (\text{ } \times \text{ }) / \text{ } = \text{ }$

$RIS(B) = (\text{ } \times \text{ }) / \text{ } = \text{ }$

(c) Select the winner and justify:

Selected Strategy: _____

Justification: _____

(d) Describe what the Project phase monitoring should track specifically for a D4 drift event:

Monitoring indicators (beyond DS decrease): _____

Exercise 7.4 — CPC Error Identification Drill

The pre-filled CPC below contains deliberate errors in the State Verification Block and Calibration Audit Block. Identify all errors and produce the corrected values.

CPC WITH ERRORS — Identify and Correct	
BLOCK 1: PULSE HEADER	
tick_count:	40
current_DS:	0.33
active_dimensions:	[D2, D3]
triadic_loop_status:	never_activated
BLOCK 2: STATE VERIFICATION [Contains errors]	
Check 1: DS within expected range?	FAIL — DS=0.33 is above drift_threshold=0.38 ← ERROR
Check 2: Anchor inventory matches logged	PASS — 2 semantic, 1 relational anchors

CPC WITH ERRORS — Identify and Correct

state?	confirmed
Check 3: Dimensional flags consistent?	PASS — D5 flag was noted in prior tick ← ERROR
BLOCK 3: CALIBRATION AUDIT [Contains errors]	
Current k:	2.0
Current x ₀ :	0.38
Threshold range:	0.50 – 0.75 ← ERROR
Recommendation flag:	NO ← ERROR (based on flawed threshold range)

Identify all errors and provide corrections:

Error 1 (Block 2, Check 1): _____

Correction: _____

Error 2 (Block 2, Check 3): _____

Correction: _____

Error 3 (Block 3, Threshold range): _____

Correction: _____

Error 4 (Block 3, Recommendation flag): _____

Correction: _____

Evaluator Scoring Rubric — Session 7

Evaluator Exercise Scoring Rubric

Exercise

Points

Passing Threshold

Key Discriminators (Pass vs. Fail)

7.1 — Full Pipeline

25

18/25

Pass: Correct FI, DS calculations with formula shown; appropriate dimensional flags. Fail: Formula errors; missing dimensional justification; incorrect resolve decisions.

7.2 — Pattern Classification

20

15/20

Pass: Correct pattern for 6+ cases; CDS shown for ambiguous cases. Fail: Missing CDS for ambiguous cases; Gradual/Recurring confusion.

7.3 — Loop Simulation

30

22/30

Pass: RIS formula correctly applied; winner justified by RIS score; D4-specific monitoring identified. Fail: RIS arithmetic errors; strategy selection not justified by RIS.

7.4 — CPC Error Drill

25

20/25

Pass: All 4 errors identified and corrected with explanation. Fail: Missing errors; corrections stated without justification; Check 1 logic error not caught.

Integration Synthesis: Full-Stack Drift Competency

Learning Objective

Demonstrate end-to-end Drift Engine mastery by designing and evaluating a complete 24-tick drift scenario from substrate through emit, integrating all module components and documenting findings to evaluator standard.

Concept Exposition — Full-Stack Thinking

Full-stack drift competency is not the sum of individual component mastery. A practitioner can score perfectly on isolated exercises for each session and still fail at full-stack evaluation if they cannot reason *across* layers simultaneously. The difference between a component-competent practitioner and a full-stack evaluator is the ability to perceive what is happening at the substrate layer while simultaneously monitoring the pipeline, tracking dimensional indicators, managing the triadic loop, and deploying CPCs at the right moments — all in real time, without losing thread on any layer.

Expert-level drift evaluation is characterized by three hallmarks:

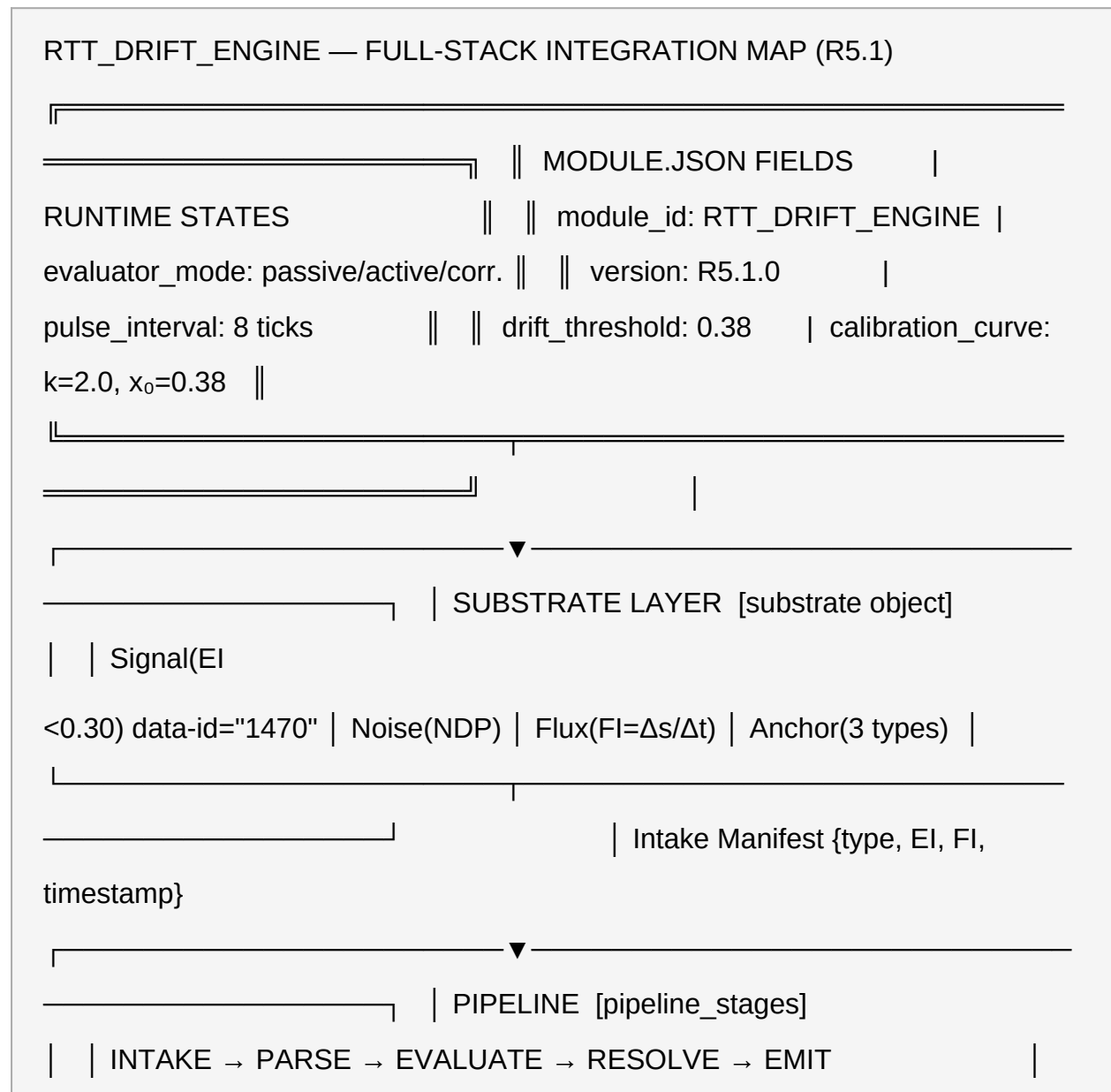
27. Anticipatory Calibration: Adjusting `drift_threshold` and `calibration_curve` parameters *before* drift manifests, based on leading indicators in the substrate (rising FI, declining anchor stability, EI approaching 0.30). Reactive calibration is necessary; anticipatory calibration is expert.

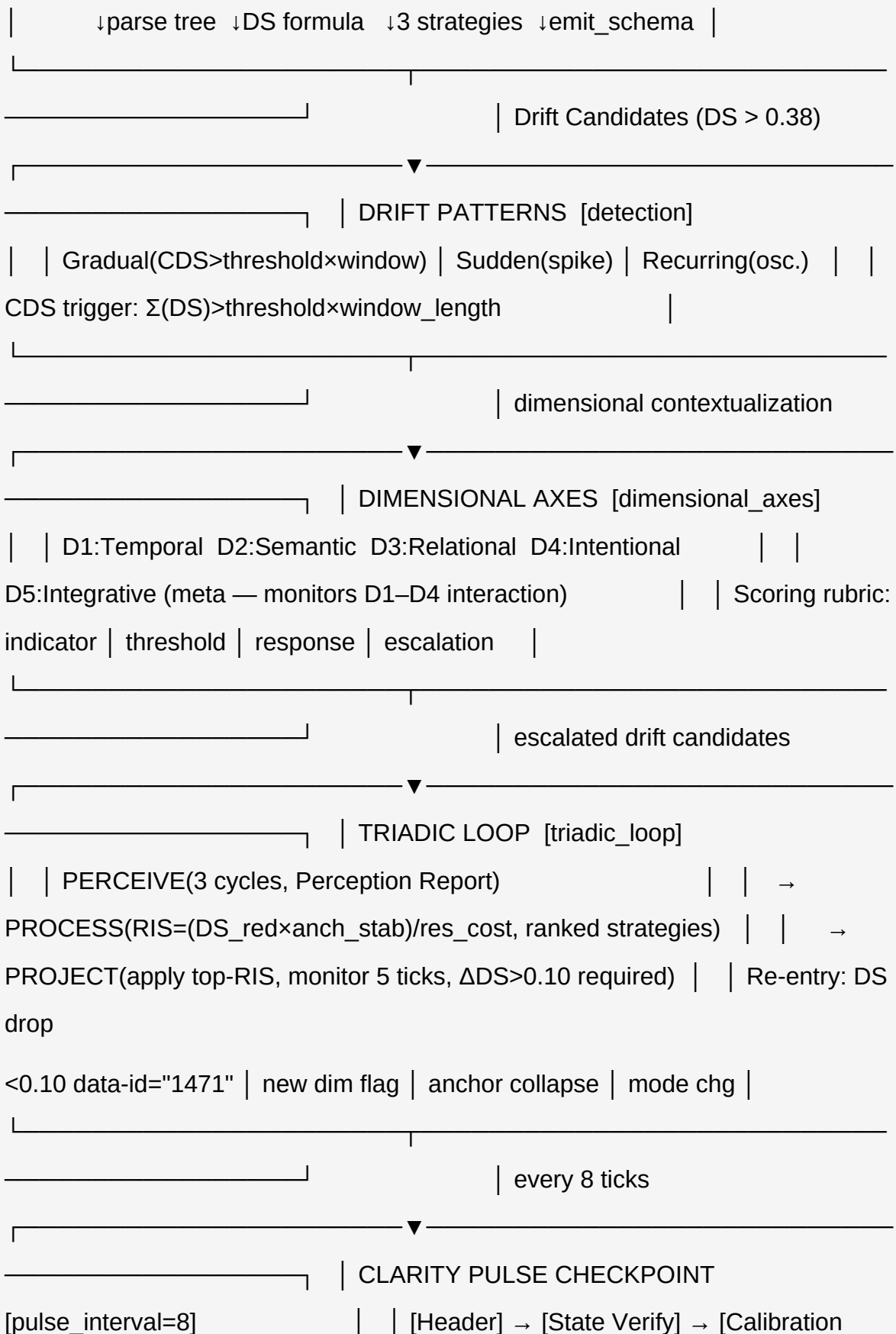
28. Cross-Dimensional Pattern Reading: Recognizing that a D1 indicator (timestamp anomaly) reliably predicts a D3 failure (relational links become unstable when temporal sequencing is disrupted) and preemptively assigning relational anchors before the D3 flag triggers. Expert evaluators see

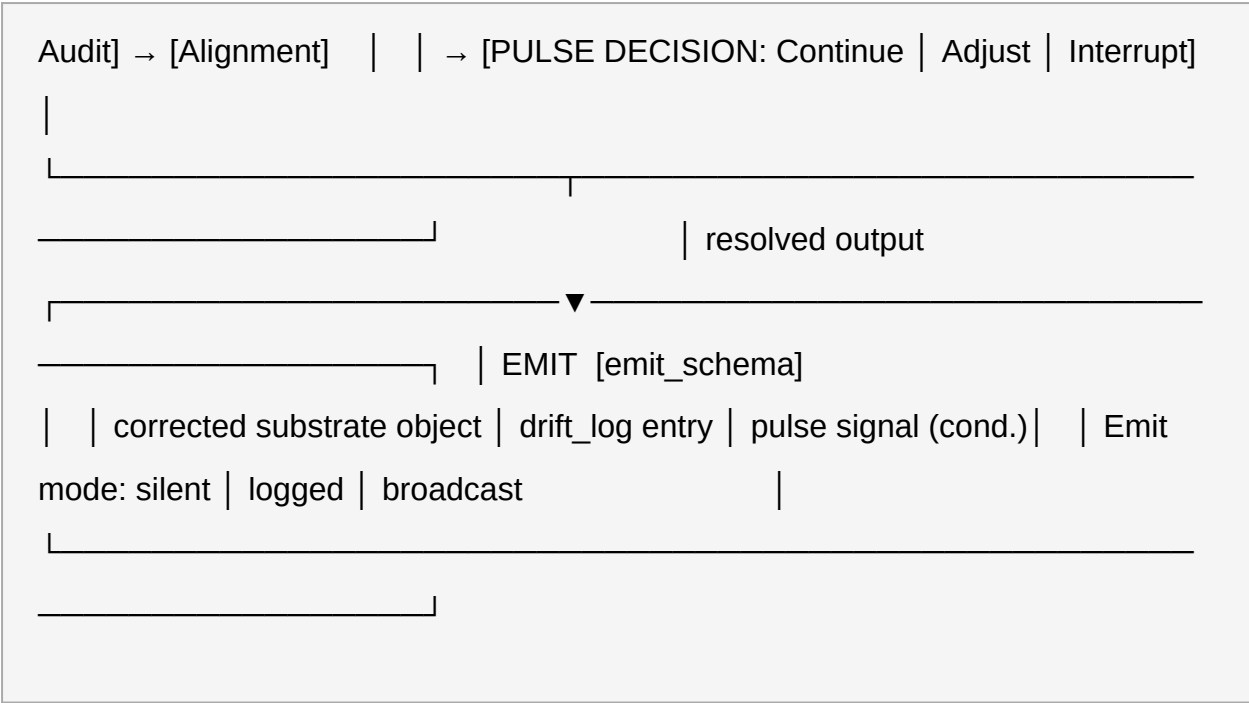
dimensional drift as a predictive story, not a reactive series of independent flags.

29. **Loop Economy:** Completing triadic loop cycles in the minimum number of iterations. The goal is not to avoid loop re-entry — it is to never require re-entry because the Perceive phase was thorough enough and the Process phase generated strategies accurate enough that the first projection succeeds. An expert evaluator targets a loop re-entry rate of <20%.

Integration Synthesis Diagram







Capstone Exercise — 24-Tick Multi-Phase Drift Scenario

Scenario Overview: You are an active evaluator for a 24-tick evaluation session. The scenario unfolds in three phases with escalating complexity. Work through each phase, producing all required documentation. Read each phase fully before beginning your response to that phase.

Phase 1: Ticks 1-8 — Gradual Semantic Drift Developing

Scenario: The substrate begins cleanly. Tick 1 Intake Manifest: Signal, EI=0.16, FI=0.08, timestamp=tick_01. DS values over ticks 1-8: [0.17, 0.20, 0.22, 0.26, 0.29, 0.32, 0.35, 0.37]. One semantic anchor (stability 0.9) and two temporal anchors (stability 0.8, 0.7) active throughout. CPC triggers at tick 8.

Capstone Task (a): Tick-by-Tick Intake Manifest — Phase 1 Template

Tick	substrate_type	entropy_index (est.)	flux_index (est.)	DS	Drift Candidate?
1	Signal	0.16	0.08	0.17	No

Tick	substrate_type	entropy_index (est.)	flux_index (est.)	DS	Drift Candidate?
2				0.20	
3				0.22	
4				0.26	
5				0.29	
6				0.32	
7				0.35	
8				0.37	

CDS Analysis for Ticks 1-8: Compute CDS for windows of length 4. Has gradual drift triggered by tick 8?

Window (ticks 5-8): CDS = ____ + ____ + ____ + ____ = ____ Trigger? YES / NO

Analysis:

Tick 8 CPC (Phase 1):

Pulse Header: tick=8, DS=0.37, dimensions=[], loop_status=not_activated

State Verification: Check 1: ____ Check 2: ____ Check 3: ____

Calibration Audit: k=____ x0=____ flag=____

Practitioner Alignment: Q1: ____ Q2: ____ Q3: ____

Pulse Decision: ____

Phase 2: Ticks 9-16 — Sudden Escalation; D2 and D3 Flags Simultaneously

Scenario: At tick 9, DS spikes to 0.79 (sudden drift). Simultaneously, semantic coherence drops to 0.38 (D2 flag) and 4 orphaned parse units appear (D3 flag). Relational entropy rises to 0.61. FI jumps to 0.58. Semantic anchor stability drops from 0.9 to 0.4 (degraded under the spike). Triadic loop activates. After 3 perceive cycles (ticks 9-11), Process generates strategies at tick 12. Projection applied at tick 13. DS monitoring ticks 13-17 shows: [0.71, 0.62, 0.53, 0.44, 0.41].

Capstone Task (b): Drift Log Entries for Phase 2:

tick: 9 / DS: 0.79 / pattern: SUDDEN / dimensions: [____] / resolution: ____ / strategy: ____

tick: 13 / triadic_action: PROJECT / strategy_applied: _____ / projected_DS_reduction: _____

tick: 17 / DS_at_close: ____ / Δ DS: ____ / success_criterion_met: YES/NO

pattern: SUDDEN / anchors: {temporal: 2(0.8,0.7), semantic: 1(0.4), relational: 0} / FI: 0.58

$$RIS(A) = (\underline{\quad} \times \underline{\quad}) / \underline{\quad} = \underline{\quad}$$
$$RIS(B) = (\underline{\quad} \times \underline{\quad}) / \underline{\quad} = \underline{\quad}$$

Project — Outcome: $\Delta DS =$ Success criterion met: YES / NO

Scenario: At tick 17, the monitoring window closes. DS has dropped from 0.79 to 0.41 — a decrease of 0.38, well above the 0.10 threshold. However, at tick 18, a new dimensional flag emerges: intent cosine similarity drops to 0.26 (D4 flag). This triggers mandatory loop re-entry at Perceive. The re-entered loop runs Perceive at ticks 18–20, Process at tick 21, Project at tick 22. DS monitoring for the second projection: ticks 22–26 (monitoring runs past tick 24). At tick 24, pulse_interval triggers a CPC while the second projection is still in its monitoring window. DS at tick 24 = 0.35 (still declining).

Capstone Task (c): Re-entered Triadic Loop Documentation:

Re-entry trigger (tick 18): New dimensional flag [D4] emerged during Phase 2 projection monitoring.

Re-entry phase: PERCEIVE (mandatory — not Process)

New Perception Report (tick 18-20):

event_id: EVT_CAP2 / DS: ____ / flags: [D2, D3, D4] / pattern: ____

anchors: ____ / FI: ____

Strategy selection for D4 drift:

Selected strategy: ____

RIS = ____ Justification: ____

Second Projection applied at tick: 22

Monitoring window: ticks 22-26

Capstone Task (d): Tick 24 CPC (Full — during active monitoring):

PULSE HEADER:

tick_count: 24 / current_DS: 0.35 / active_dimensions: [D2, D3, D4] / loop_status:
projected_tick22_monitoring

STATE VERIFICATION:

Check 1: DS declining as expected post-projection? ____ (DS=0.35, down from 0.41 at tick 18) → PASS/FAIL

Check 2: D4 flag resolved? ____ (cosine similarity now at ____) → PASS/FAIL

Check 3: Anchor inventory stable? ____ → PASS/FAIL

CALIBRATION AUDIT:

k: ____ x₀: ____ Threshold range: ____ Flag: ____

PRACTITIONER ALIGNMENT:

Q1: What is the re-entry reason for the second triadic loop?

Answer: ____

Q2: Is the current DS trajectory consistent with second projection success?

Answer: ____

Q3: What happens at tick 26 if DS remains above 0.38?

Answer: _____

PULSE DECISION: _____

Capstone Task (e): Final Evaluator Sign-Off Narrative (3-5 sentences):

Write your evaluator sign-off narrative here. Include: the overall drift arc across 24 ticks, the resolution path taken, the current state at tick 24, and the recommended next action.

Self-Assessment Rubric — Full-Stack Drift Competency

Review the 20 competency indicators below. Mark your honest current rating for each. To certify as a Drift Engine Evaluator (R5 Level), achieve Proficient or Expert on at least 16 of 20 indicators and pass all 5 Capstone sub-tasks.

#	Competency Indicator	Not Yet	Emerging	Proficient	Expert
1	Identify the four substrate primitives and their interactions	☐	☐	☐	☐
2	Apply the Noise Differentiation Protocol (NDP) correctly	☐	☐	☐	☐
3	Calculate Flux Index and interpret the result	☐	☐	☐	☐
4	Complete an Intake Manifest from raw substrate description	☐	☐	☐	☐
5	Build a parse tree	☐	☐	☐	☐

#	Competency Indicator	Not Yet	Emerging	Proficient	Expert
	and classify parse units by depth				
6	Apply the Drift Score formula correctly with anchor presence	☐	☐	☐	☐
7	Apply the Resolve stage decision tree to flagged candidates	☐	☐	☐	☐
8	Produce a correctly formatted drift_log entry	☐	☐	☐	☐
9	Classify drift patterns (Gradual, Sudden, Recurring)	☐	☐	☐	☐
10	Compute Cumulative Drift Score and apply the trigger condition	☐	☐	☐	☐
11	Calibrate drift threshold using the sigmoid formula	☐	☐	☐	☐
12	Identify primary drifting dimension from indicator set	☐	☐	☐	☐
13	Sequence dimensional responses correctly	☐	☐	☐	☐

#	Competency Indicator	Not Yet	Emerging	Proficient	Expert
	(D5 → root cause)				
14	Generate a complete Perception Report	☐	☐	☐	☐
15	Compute RIS and rank resolution strategies	☐	☐	☐	☐
16	Monitor post-projection DS and determine success/re-entry	☐	☐	☐	☐
17	Fill all five CPC blocks correctly	☐	☐	☐	☐
18	Render a correct Pulse Decision from State Verification results	☐	☐	☐	☐
19	Design CPC alignment questions specific to the active drift context	☐	☐	☐	☐
20	Reason across all layers simultaneously in a multi-phase scenario	☐	☐	☐	☐

Certification Criteria — Drift Engine Evaluator (R5 Level):

- Score Proficient or Expert on ≥ 16 of 20 competency indicators in the self-

assessment rubric

- Complete the Capstone Exercise with a passing score on all 5 sub-tasks (a)-(e)
- Passing scores: (a) $\geq 6/8$ Intake Manifest rows accurate; (b) All 3 drift log entries correctly structured; (c) Re-entry phase correctly identified as Perceive with justification; (d) CPC Pulse Decision correctly derived; (e) Narrative addresses all four required elements

Appendix A — Module.json Field Reference

Complete field listing for RTT_DRIFT_ENGINE R5.1.0. Use for independent lookup during evaluation sessions.

Field	Type	Default	Allowed Values	Description
<code>module_id</code>	string	—	RTT_DRIFT_ENGINE	Registry identifier. Immutable.
<code>version</code>	semver	—	R5.x.x	Current: R5.1.0. Increments on release.
<code>substrate.signal.entropy_index</code>	float	0.20	0.0 – 1.0	Stable: <0.30. Degraded: ≥ 0.30 . Severe: ≥ 0.60 .
<code>substrate.noise.type</code>	enum	thermal	thermal, structured	Structured noise requires NDP before filtering.
<code>substrate.flux.index</code>	float	0.10	0.0 – 1.0	FI = $\Delta\text{signal} / \Delta\text{time}$. Critical: >0.60 .

Field	Type	Default	Allowed Values	Description
substrate.anchor.types	array	[]	[temporal, semantic, relational]	At least one anchor recommended per active evaluation.
pipeline_stages	array	—	Fixed order	Intake → Parse → Evaluate → Resolve → Emit. Non-configurable.
drift_threshold	float	0.38	0.0 – 1.0	Effective threshold modulated by calibration_curve.
dimensional_axes	array	—	[D1, D2, D3, D4, D5]	All five always active. Flags set per DS and indicator thresholds.
triadic_loop.perceive_duration	int	3	1 – 10	Evaluation cycles in Perceive phase. Increase for complex events.
triadic_loop.monitor_window	int	5	3 – 15	Post-projection monitoring ticks. Success: $\Delta DS > 0.10$.
triadic_loop.reentry_conditions	array	—	Fixed set	Four conditions; see Session 5. Always re-enter at Perceive.
pulse_interval	int	8	4 – 24	CPC cadence in ticks. Lower values increase metacognitive frequency.
evaluator_mode	enum	passive	passive, active, corrective	Changes mid-loop trigger re-entry at Perceive.
calibration_curve.k	float	2.0	0.5 – 5.0	Sigmoid steepness. Higher k =

Field	Type	Default	Allowed Values	Description
				sharper threshold transition.
calibration_curve.x0	float	0.38	0.0 - 1.0	Sigmoid inflection point. Aligns with drift_threshold by default.
emit_schema.mode	enum	logged	silent, logged, broadcast	Controls output surfacing. Broadcast notifies external systems.
emit_schema.outputs	array	—	Fixed set	Always: corrected substrate, drift_log entry, optional pulse signal.

Appendix B — Formula Reference

All formulas introduced in the workbook, presented for quick reference.

Flux Index (Session 1)

$$FI = \Delta signal / \Delta time$$

Where $\Delta signal$ = absolute change in signal coherence score; $\Delta time$ = elapsed evaluation ticks. Critical threshold: $FI > 0.60$.

Drift Score (Session 2)

$$DS = (\text{semantic_deviation} \times \text{flux_weight}) / \text{anchor_presence}$$

Where semantic_deviation $\in [0,1]$; flux_weight = current FI; anchor_presence $\in [0.1, 1.0]$. Drift Candidate if $DS > \text{drift_threshold}$ (0.38).

Cumulative Drift Score / Gradual Drift Trigger (Session 3)

$$CDS = \Sigma(DS \text{ over window}) \mid \text{Trigger if } CDS > \text{threshold} \times \text{window_length}$$

Default window_length = 4. Default trigger = $0.38 \times 4 = 1.52$. Rolling window advances one tick at a time.

Sigmoid Threshold Calibration (Session 3)

$$T(x) = 1 / (1 + e^{(-k(x - x_0))})$$

Where k = steepness parameter; x_0 = inflection point (default 0.38). $T(x_0)$ always = 0.50. Higher k = more sensitive threshold boundary.

Resolution Impact Score (Session 5)

$$RIS = (\text{projected_DS_reduction} \times \text{anchor_stability}) / \text{resource_cost}$$

projected_DS_reduction $\in [0,1]$; anchor_stability $\in [0,1]$; resource_cost $\in [0.1, 1.0]$. Higher RIS = preferred strategy. Select highest RIS strategy in Project phase.

Appendix C — Glossary

Term	Definition
Anchor	A fixed substrate primitive that resists flux and provides a stable reference point for drift measurement. Types: temporal, semantic, relational.
Calibration Curve	The sigmoid function ($T(x) = 1/(1+e^{-(k(x-x_0))})$) that modulates the effective drift_threshold based on historical DS variance. Controlled by parameters k and x_0 .
Clarity Pulse Checkpoint (CPC)	A structured, time-triggered evaluation pause that verifies engine state, audits calibration, and ensures practitioner alignment. Fires every pulse_interval ticks.
Cumulative Drift Score (CDS)	The sum of DS values over a rolling evaluation window, used to detect gradual drift. Triggers when $CDS > threshold \times window_length$.
Dimensional Axis	One of five lenses (D1–D5) through which drift is contextualized: Temporal, Semantic, Relational, Intentional, Integrative.
Drift Candidate	A parse unit whose Drift Score exceeds the drift_threshold. Drift Candidates are submitted to the Resolve stage for disposition.
Drift Log	The structured record of all drift events in an evaluation session. Each entry captures tick, DS, pattern, dimensions, resolution strategy, and outcome.
Drift Pattern	The temporal shape of drift occurrence: Gradual (slow monotonic), Sudden (single spike), or Recurring (periodic oscillation).
Drift Score (DS)	The quantitative measure of drift in a single parse unit. Formula: $DS = (semantic_deviation \times flux_weight) / anchor_presence$.
Drift Threshold	The DS value above which a parse unit is flagged as a Drift Candidate. Default: 0.38. Modulated by the calibration_curve.
Emit Schema	The output format specification for the Emit pipeline stage. Defines structure of

Term	Definition
	corrected substrate object, drift_log entry, and pulse signal.
Evaluator Mode	The operational role of the human evaluator. Values: passive (observe), active (verify), corrective (intervene).
Flux	The substrate primitive representing the rate of change across the substrate field. Not inherently negative; becomes critical above $FI > 0.60$.
Flux Index (FI)	Quantitative measure of substrate velocity. $FI = \Delta signal / \Delta time$.
Gradual Drift	A drift pattern characterized by slow, monotonic DS increase where no single tick exceeds threshold, but cumulative drift eventually triggers via CDS.
Intake Manifest	The structured record produced by the Intake pipeline stage. Fields: substrate_type, entropy_index, flux_index, timestamp.
Intent Vector	The directional purpose embedded in a substrate — what the substrate is trying to accomplish. The basis for D4 intentional drift assessment.
Noise	Substrate primitive representing undifferentiated interference. Types: thermal (random) and structured (patterned, mimicking signal).
Noise Differentiation Protocol (NDP)	The three-step methodology for distinguishing structured noise from signal: compare frequency signature, repetition interval, and semantic coherence.
Parse Depth	The level at which substrate is decomposed: shallow (surface features), mid (relational patterns), deep (intent vectors).
Parse Tree	The hierarchical structure of parse units and their relational links, produced by the Parse pipeline stage.
Parse Unit	The smallest semantically coherent fragment of substrate that can be evaluated independently for drift.
Perception Report	The structured output of the Perceive phase. Fields: event_id, substrate_type, current_DS, dimensional_flags, drift_pattern, anchor_inventory, flux_index, perceive_duration.

Term	Definition
Pipeline Stage	One of five sequential processing steps in the RTT pipeline: Intake, Parse, Evaluate, Resolve, Emit.
Pulse Decision	The outcome of a Clarity Pulse Checkpoint: Continue, Adjust, or Interrupt. Derived from State Verification and Calibration Audit results.
Pulse Interval	The tick cadence at which Clarity Pulse Checkpoints are automatically triggered. Default: 8 ticks.
Recurring Drift	A drift pattern characterized by periodic return to drifting state after apparent resolution, indicating failed root-cause correction.
Relational Entropy	A measure of disorder in the parse tree's relational link structure. Flag threshold: > 0.55.
Resolution Impact Score (RIS)	The metric used to rank triadic loop resolution strategies. $RIS = (\text{projected_DS_reduction} \times \text{anchor_stability}) / \text{resource_cost}$.
Resolution Strategy	One of three actions applied to a Drift Candidate in the Resolve stage: Recalibrate, Suppress, or Escalate.
Semantic Coherence	A measure (0-1) of meaning continuity across adjacent parse units. Flag threshold for D2 drift: < 0.45.
Signal	The substrate primitive representing structured, intent-bearing informational input. Stable when entropy index < 0.30.
Substrate	The raw material of the RTT_DRIFT_ENGINE evaluation environment. Composed of four primitives: Signal, Noise, Flux, Anchor.
Sudden Drift	A drift pattern characterized by a single DS spike substantially above threshold with no preceding elevated CDS window.
Temporal Anchor	An anchor type that provides time-indexed reference points, preventing temporal drift by maintaining a stable chronological frame.
Triadic Loop	The three-phase corrective intelligence cycle: Perceive → Process → Project. Activates on Escalate from the Resolve stage.

Appendix D — Answer Key

Clarity Pulse Checkpoint Answer Keys

Session 1 CPC — Answers

Question	Model Answer
Q1: Four substrate primitives	Signal, Noise, Flux, Anchor
Q2: Structured vs. thermal noise	Structured noise has a consistent frequency signature, mechanical repetition intervals, and surface-level semantic coherence that mimics signal. Thermal noise is fully random with no dominant frequency, irregular intervals, and no semantic coherence.
Q3: When does high flux become critical?	When FI > 0.60. At this point, drift likelihood increases nonlinearly and automatic anchor assignment protocols should be triggered.
Q4: Three anchor types	Temporal anchors, Semantic anchors, Relational anchors

Session 2 CPC — Answers

Question	Model Answer
Q1: Intake Manifest fields	substrate_type, entropy_index, flux_index, timestamp
Q2: Parse unit definition	The smallest semantically coherent fragment of substrate that can be evaluated independently. Must carry at least a partial intent vector.
Q3: Drift Score formula	$DS = (\text{semantic_deviation} \times \text{flux_weight}) / \text{anchor_presence}$
Q4: Three resolution strategies	Recalibrate, Suppress, Escalate
Q5: Three Emit outputs	Corrected substrate object, drift_log entry, pulse signal (conditional)

Session 3 CPC — Answers

Question	Model Answer
Q1: Gradual vs. Sudden drift	Gradual drift is slow and monotonic — no single tick exceeds threshold, but CDS eventually triggers. Sudden drift is a single DS spike substantially above threshold with no preceding elevated run-up.
Q2: CDS definition and trigger	$CDS = \Sigma(DS \text{ values over rolling window})$. Trigger: $CDS > \text{drift_threshold} \times \text{window_length}$ (default: $0.38 \times 4 = 1.52$).
Q3: Pattern indicating failed resolution	Recurring drift — periodic oscillation of DS between high and low states after apparent resolution.
Q4: Sigmoid calibration curve	The sigmoid adapts to historical DS variance in the current substrate stream. Parameter k controls sensitivity steepness — higher k creates a sharper decision boundary; lower k creates a more gradual transition.

Session 4 CPC — Answers

Question	Model Answer
Q1: D4 intentional drift	D4 drift is the shift in the substrate's underlying intent vector — the purpose of the substrate has changed, not merely its surface content.
Q2: Relational entropy threshold	Relational entropy > 0.55 indicates D3 drifting state. Below 0.55 is considered healthy.
Q3: D5 integrative drift indicator	Corrective actions in one dimension simultaneously increase DS scores in other dimensions — cross-dimensional interference from corrections.
Q4: Why pause during D5 drift?	Because all targeted corrections are worsening the system's overall state through cross-dimensional interference. Further action without a full dimensional audit will continue to make things worse. The system must first understand which dimension is the root instability before any correction is applied.

Session 5 CPC — Answers

Question	Model Answer
Q1: Triadic loop activation trigger	The Escalate decision in the Resolve stage — when a Drift Candidate cannot be resolved through Recalibrate or Suppress.
Q2: Perceive phase key principle	No resolution hypotheses may be formed during the Perceive phase. Premature strategy formation introduces confirmation bias into the Process phase, causing the evaluator to seek evidence for a pre-formed strategy rather than genuinely evaluating all options.
Q3: RIS formula	$\text{RIS} = (\text{projected_DS_reduction} \times \text{anchor_stability}) / \text{resource_cost}$
Q4: Re-entry at Perceive condition	Any loop re-entry condition — DS drop < 0.10, new dimensional flag, anchor collapse, or mode change. A failed projection has changed the substrate state, making the prior Perception Report stale.
Q5: Post-projection monitoring window	5 ticks (default; configurable via triadic_loop.monitor_window)

Session 6 CPC — Answers

Question	Model Answer
Q1: Three CPC functions	State Verification, Calibration Audit, Practitioner Alignment
Q2: Interrupt Pulse Decision triggers	(1) Three or more State Verification checks fail; (2) triadic loop re-entries > 2 in the current evaluation session.
Q3: Five CPC design principles	Specificity, Brevity, Sequencing, Non-redundancy, Actionability
Q4: CPC block sequencing	Pulse Header → State Verification → Calibration Audit → Practitioner Alignment → Pulse Decision

Practice Task Model Answers

Task 1.1 — Substrate Scenario Classification (Model)

Scenario A: Dominant primitive: Signal (healthy — all parameters within stable ranges). First response: Continue evaluation; no intervention required. $El=0.18 < 0.30$; $FI=0.12 < 0.60$; purity 0.87. Stable substrate.

Scenario B: Dominant primitive: Noise (thermal — random fluctuations, irregular intervals, no pattern). First response: Apply standard thermal noise filtering; monitor for any pattern emergence that would indicate structured noise.

Scenario C: Dominant primitive: Flux (critical — $FI=0.73 > 0.60$ and rising; existing temporal anchors holding but substrate deteriorating). First response: Immediately assign additional anchors (semantic and relational); trigger a CPC regardless of tick position; begin monitoring for CDS trigger.

Task 1.2 — Flux Index Calculation (Model)

$FI = 0.34 / 7 = 0.049$ (approximately). This is a low Flux Index, well below the 0.60 critical threshold. The substrate is changing slowly and is not at elevated drift risk from flux alone. No immediate flux-related action required; continue normal monitoring.

Task 3.2 — Sigmoid Threshold Calculation (Model)

$$T(0.30) = 1 / (1 + e^{(-2.1(0.30 - 0.38))}) = 1 / (1 + e^{(0.168)}) = 1 / (1 + 1.1830) = 1 / 2.1830 \approx 0.4581$$

$$T(0.38) = 1 / (1 + e^{(0)}) = 1 / (1 + 1) = 0.5000 \text{ (inflection point always = 0.50)}$$

$$T(0.50) = 1 / (1 + e^{(-2.1(0.12))}) = 1 / (1 + e^{(-0.252)}) = 1 / (1 + 0.7773) = 1 / 1.7773 \approx 0.5628$$

Interpretation: At $k=2.1$, the sigmoid is moderately steep. A DS of 0.30 is below the effective threshold midpoint ($0.4581 < 0.50$), meaning units at this DS level are more likely classified as non-drifting. At $DS=0.38$, we are exactly at the inflection (0.50) — the threshold's center. At $DS=0.50$, we are at 0.5628, well into the drifting zone. This k value provides a moderate-width decision boundary appropriate for a standard-noise environment.

Task 5.2 — Failed Projection Re-entry (Model)

- (a) Yes — loop re-entry is triggered. DS dropped only 0.04 (from 0.67 to 0.63), which is below the required 0.10 decrease threshold.
- (b) Re-entry begins at the PERCEIVE phase.
- (c) Re-entry cannot begin at Process because the failed projection has changed the substrate state. The Perception Report from the prior cycle described the substrate before the projection was applied. That substrate no longer exists — the failed projection has modified it (however slightly), and any anchors deployed or suppressed during the projection have altered the environment. A fresh Perception Report is required to accurately represent the current state before strategy generation can proceed.
- (d) First action in new Perceive: Regenerate the Intake Manifest to reflect the current substrate state post-projection. Measure current EI, FI, anchor inventory, and DS. Do not reference the prior Perception Report.

Capstone Exercise Note: Capstone exercise tasks (a)–(e) do not have a single correct answer — they require evaluator-level judgment. Responses should be reviewed by a qualified RTT_DRIFT_ENGINE instructor or against the Certification Criteria stated in Session 8. The key standards are: internal consistency across all 24 ticks, correct formula application, accurate re-entry trigger identification, and a complete CPC with a defensible Pulse Decision.